Binary Relations

Discrete Math, Fall 2025

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Relations

"In mathematics you don't understand things. You just get used to them."

— John von Neumann

René Descartes

Évariste Galois

Ernst Schröder

Michael Rabin

Herbert Wilf

Relations as Sets

Definition 1: A *binary relation* R on sets A and B is a subset of the Cartesian product $A \times B$.

Notation: If $R \subseteq A \times B$, we write "$a\ R\ b$" to mean that element $a \in A$ is *related* to element $b \in B$.

Formally, $a\ R\ b$ iff $\langle a, b \rangle \in R$.

Note: R is used to denote both the relation itself ($a\ R\ b$) *and* the set of pairs ($R \subseteq A \times B$).

Note: The *order* of elements in the pair *matters*: $\langle a, b \rangle \in R$ denotes that a is related to b, not the other way around, unless there is *another* pair $\langle b, a \rangle$ in the relation.

Example: $R = \{\langle n, k \rangle \mid n, k \in \mathbb{N} \text{ and } n < k\}$

Definition 2:

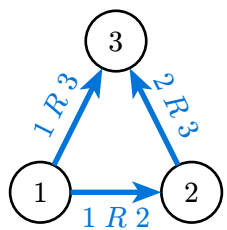
- A binary relation $R \subseteq A \times B$ on two different sets A and B is called *heterogeneous*.
- A binary relation $R \subseteq M^2$ on the same set M is called *homogeneous*.

Graph Representation

Definition 3: A homogeneous relation $R \subseteq M^2$ can be represented as a *directed graph* where:

- Vertices correspond to elements of M
- There is a directed edge from x to y if $x \; R \; y$, *i.e.* $\langle x, y \rangle \in R$

Example: For $M = \{1, 2, 3\}$ and $R = \{\langle 1, 2 \rangle, \langle 2, 3 \rangle, \langle 1, 3 \rangle\}$, the graph has vertices $\{1, 2, 3\}$ and directed edges $1 \to 2$, $2 \to 3$, and $1 \to 3$.

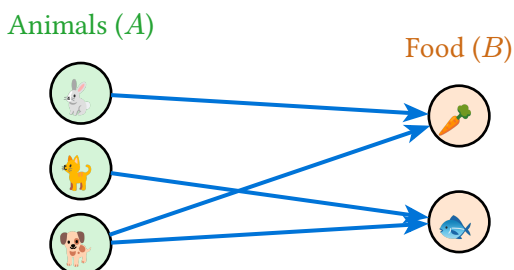


Graph Representation [2]

Definition 4: A heterogeneous relation $R \subseteq A \times B$ can be represented as a *bipartite graph* where:

- Vertices in one partition correspond to elements of A
- Vertices in the other partition correspond to elements of B
- There is a directed edge from $a \in A$ to $b \in B$ if $a \; R \; b$, *i.e.* $\langle a, b \rangle \in R$

Example: For animals $A = \{$🐰, 🐈, 🐕$\}$, food $B = \{$🥕, 🐟$\}$, and relation R = "likes to eat", we have the bipartite graph with animal vertices on the left side and food vertices on the right side with four edges.



Matrix Representation

Definition 5: A binary relation $R \subseteq A \times B$ can be represented as a *matrix* $M_R = [\![R]\!]$ where:

- Rows correspond to elements of A
- Columns correspond to elements of B
- $M_R[i, j] = 1$ if $a_i \; R \; b_j$, and $M_R[i, j] = 0$ otherwise

Example: Let $A = \{a, b, c\}$, $B = \{x, y\}$, and $R = \{\langle a, x\rangle, \langle b, x\rangle, \langle c, y\rangle\}$. The matrix representation is:

$$[\![R]\!] = \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{where rows are } \{a, b, c\} \text{ and columns are } \{x, y\}$$

Special Relations

Definition 6: For any set M, we define these special relations:

- *Empty relation*: $\emptyset \subseteq M^2$ (no elements are related)
- *Identity relation*: $I_M = \{\langle x, x\rangle \mid x \in M\}$ (each element related only to itself)
- *Universal relation*: $U_M = M^2$ (every element related to every element)

Example: For $M = \{a, b, c\}$:

- Empty: $\emptyset$
- Identity: $\{\langle a, a\rangle, \langle b, b\rangle, (c, c)\}$
- Universal: $\{\langle a, a\rangle, \langle a, b\rangle, \langle a, c\rangle \langle b, a\rangle, \langle b, b\rangle, \langle b, c\rangle, \langle c, a\rangle, \langle c, b\rangle, \langle c, c\rangle\}$ (all 9 pairs)

Operations on Relations

Definition 7: For relations $R, S \subseteq A \times B$:

- *Union*: $R \cup S = \{\langle a, b\rangle \mid \langle a, b\rangle \in R \text{ or } \langle a, b\rangle \in S\}$
- *Intersection*: $R \cap S = \{\langle a, b\rangle \mid \langle a, b\rangle \in R \text{ and } \langle a, b\rangle \in S\}$
- *Complement*: $\overline{R} = (A \times B) \setminus R$

Definition 8: For a relation $R \subseteq A \times B$, the *inverse* (or *converse*, or *dual*) relation is:

$$R^{-1} = \{\langle b, a\rangle \mid \langle a, b\rangle \in R\} \subseteq B \times A$$

Note: Other notations include R^T (transpose), R^C (converse), R° (reciprocal).

Example: If $R = \{\langle 1, x\rangle, \langle 2, y\rangle, \langle 2, z\rangle\}$, then $R^{-1} = \{\langle x, 1\rangle, \langle y, 2\rangle, \langle z, 2\rangle\}$.

Example: For the usual order relations, the converse is the naively "opposite" order:

- The converse of "less" is "greater": $<^T = >$
- The converse of "less or equal" is "greater or equal": $\leq^T = \geq$

Properties of Relations

Properties of Homogeneous Relations

Definition 9: A relation $R \subseteq M^2$ is *reflexive* if every element is related to itself:

$$\forall x \in M. (x \; R \; x)$$

Definition 10: A relation $R \subseteq M^2$ is *symmetric* if for every pair of elements, if one is related to the other, then the reverse is also true:

$$\forall x, y \in M. (x \; R \; y) \to (y \; R \; x)$$

Definition 11: A relation $R \subseteq M^2$ is *transitive* if for every three elements, if the first is related to the second, and the second is related to the third, then the first is also related to the third:

$$\forall x, y, z \in M. (x \; R \; y \land y \; R \; z) \to (x \; R \; z)$$

More Properties

Definition 12: A relation $R \subseteq M^2$ is *irreflexive* if no element is related to itself:

$$\forall x \in M. (x \not R\, x)$$

Definition 13: A relation $R \subseteq M^2$ is *antisymmetric* if for every pair of elements, if both are related to each other, then they must be equal:

$$\forall x, y \in M. (x\ R\ y \wedge y\ R\ x) \rightarrow (x = y)$$

Definition 14: A relation $R \subseteq M^2$ is *asymmetric* if for every pair of elements, if one is related to the other, then the reverse is not true:

$$\forall x, y \in M. (x\ R\ y) \rightarrow (y \not R\, x)$$

Note: *irreflexive* + *antisymmetric* = *asymmetric*.

Notes on Properties

- Reflexivity and irreflexivity are *not* mutually exclusive if $M = \emptyset$ (both are *vacuously*[1] `true`).
- Symmetry and antisymmetry are *not* mutually exclusive (*e.g.* identity relation).
- Asymmetry implies irreflexivity and antisymmetry.

[1] A statement "for all x in emptyset, $P(x)$" is considered `true` because there are *no counterexamples* in the empty set.

Additional Properties

Definition 15: A relation $R \subseteq M^2$ is:

- *Coreflexive* if $R \subseteq I_M$ (only related to themselves, if at all):

$$\forall x, y \in M. (x \; R \; y) \to (x = y)$$

- *Right Euclidean* if whenever an element is related to two others, those two are related:

$$\forall x, y, z \in M. (x \; R \; y \land x \; R \; z) \to (y \; R \; z)$$

- *Left Euclidean* if whenever two elements are both related to a third, they are related to each other:

$$\forall x, y, z \in M. (y \; R \; x \land z \; R \; x) \to (y \; R \; z)$$

Examples:

- Identity relation I_M is coreflexive. Any subset of I_M is also coreflexive.
- Equality relation "$=$" is left and right Euclidean.
- "Being in the same equivalence class" is Euclidean in both directions.

Equivalence Relations

Equivalence Relations

Definition 16: A relation $R \subseteq M^2$ is an *equivalence relation* if it is reflexive, symmetric and transitive.

Example (Equality): The *identity relation* $I_M = \{\langle x, x\rangle \mid x \in M\}$ is an equivalence relation on any set M.

Note: The identity relation is just the common equality relation "$=$".

Verification:

- **Reflexive:** $x \; I_M \; x$ for all $x \in M$ (by definition of I_M) ✓
- **Symmetric:** If $x \; I_M \; y$, then $x = y$, thus $y = x$, so $y \; I_M \; x$ ✓
- **Transitive:** If $x \; I_M \; y$ and $y \; I_M \; z$, then $x = y = z$, so $x \; I_M \; z$ ✓

This is the "finest" possible equivalence relation — it distinguishes every element.

Equivalence Classes

Definition 17: Let $R \subseteq M^2$ be an equivalence relation on a set M. The *equivalence class* of an element $x \in M$ under R is the set of all elements related to x:

$$[x]_R = \{y \in M \mid x \; R \; y\}$$

Example (Equality): For the identity relation I_M on set $M = \{a, b, c\}$:

- $[a]_{I_M} = \{a\}$ (only a is equal to a)
- $[b]_{I_M} = \{b\}$ (only b is equal to b)
- $[c]_{I_M} = \{c\}$ (only c is equal to c)

Each element forms its own equivalence class under equality.

Examples of Equivalence Relations

Example (Modular arithmetic): For any positive integer n, *congruence modulo n* on $\mathbb{Z}$ is defined by:

$$a \equiv b \pmod{n} \quad \text{iff} \quad n \mid (b - a)$$

Verification:

- **Reflexive:** $a \equiv a \pmod{n}$ since $n \mid (a - a) \iff n \mid 0$ ✓
- **Symmetric:** If $a \equiv b \pmod{n}$, then $n \mid (b - a) \iff n \mid (a - b)$, thus $b \equiv a \pmod{n}$ ✓
- **Transitive:** If $a \equiv b \pmod{n}$ and $b \equiv c \pmod{n}$, then $n \mid (b - a)$ and $n \mid (c - b)$, so $n \mid ((c - b) + (b - a)) \iff n \mid (c - a)$, thus $a \equiv c \pmod{n}$ ✓

Equivalence classes (remainders): Let $n = 5$ and $M = \{0, 1, \ldots, 9\}$.

- $[0]_{\equiv} = \{0, 5\} = \{x \in M \mid x \equiv 0 \pmod{5}\}$
- $[1]_{\equiv} = \{1, 6\} = \{x \in M \mid x \equiv 1 \pmod{5}\}$
- $[2]_{\equiv} = \{2, 7\} = \{x \in M \mid x \equiv 2 \pmod{5}\}$
- $[3]_{\equiv} = \{3, 8\} = \{x \in M \mid x \equiv 3 \pmod{5}\}$
- $[4]_{\equiv} = \{4, 9\} = \{x \in M \mid x \equiv 4 \pmod{5}\}$

Examples of Equivalence Relations [2]

Example (Same absolute value): On $M = \{-3, -2, -1, 0, 1, 2, 3\}$, define relation R by:

$$x \; R \; y \quad \text{iff} \quad |x| = |y|$$

Verification:

- **Reflexive:** $|x| = |x|$ for all x ✓
- **Symmetric:** If $|x| = |y|$, then $|y| = |x|$ ✓
- **Transitive:** If $|x| = |y|$ and $|y| = |z|$, then $|x| = |z|$ ✓

Equivalence classes:

- $[0]_R = \{0\}$
- $[1]_R = [-1]_R = \{-1, 1\}$
- $[2]_R = [-2]_R = \{-2, 2\}$
- $[3]_R = [-3]_R = \{-3, 3\}$

Each positive number is equivalent to its negative counterpart.

Examples of Equivalence Relations [3]

Example (Same string length): On the set of all finite strings Σ^* over alphabet Σ, define:

$$s_1 \; R \; s_2 \quad \text{iff} \quad |s_1| = |s_2|$$

where $|s|$ denotes the length of string s.

Verification:

- **Reflexive:** Every string has the same length as itself ✓
- **Symmetric:** If two strings have the same length, the relation is symmetric ✓
- **Transitive:** If $|s_1| = |s_2|$ and $|s_2| = |s_3|$, then $|s_1| = |s_3|$ ✓

Equivalence classes: $[s]_R = \{t \in \Sigma^* \mid |t| = |s|\}$

For example, over $\Sigma = \{a, b\}$:

- $[\varepsilon]_R = \{\varepsilon\}$ (empty string)
- $[\mathrm{a}]_R = \{\mathrm{a}, \mathrm{b}\}$ (all strings of length 1)
- $[\mathrm{ab}]_R = \{\mathrm{aa}, \mathrm{ab}, \mathrm{ba}, \mathrm{bb}\}$ (all strings of length 2)

Examples of Equivalence Relations [4]

Example (Living in the same city): Let P be the set of people, and define relation R by:

$$p_1 \; R \; p_2 \quad \text{iff} \quad p_1 \text{ and } p_2 \text{ live in the same city}$$

Verification:

- **Reflexive:** Every person lives in the same city as themselves ✓
- **Symmetric:** If person A and B live in the same city, then B and A live in the same city ✓
- **Transitive:** If A and B live in the same city, and B and C do so, then A and C live in the same city ✓

Equivalence classes: Each equivalence class consists of all people living in the same city.

- $[\text{Alice}]_R$ = all people living in Alice's city
- This naturally partitions the population by cities

Application: This relation captures a common social grouping based on location.

Examples of Equivalence Relations [5]

Example (Similarity of triangles): Let T be the set of all triangles in the plane. Define relation $\sim$ by:

$$\triangle_1 \sim \triangle_2 \quad \text{iff} \quad \triangle_1 \text{ and } \triangle_2 \text{ are similar}^2$$

Verification:

- **Reflexive:** Every triangle is similar to itself ✓
- **Symmetric:** If triangle A is similar to triangle B, then triangle B is similar to triangle A ✓
- **Transitive:** If $A \sim B$ and $B \sim C$, then $A \sim C$ (similarity is transitive) ✓

Equivalence classes: Each class consists of all triangles with the same shape (but possibly different sizes).

- All equilateral triangles form one equivalence class
- All right triangles with legs in ratio 3:4:5 form another equivalence class

Geometric significance: This relation captures the concept of "same shape, different size."

[2] Two triangles are similar if their corresponding angles are equal.

Examples of Equivalence Relations [6]

Example (Rational numbers): On the set of ordered pairs $\mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$, define:

$$\langle a, b \rangle \sim \langle c, d \rangle \quad \text{iff} \quad a \cdot d = b \cdot c$$

This relation is used to construct rational numbers $\mathbb{Q}$ from integer pairs.

Verification:

- **Reflexive:** $\langle a, b \rangle \sim \langle a, b \rangle$ since $a \cdot b = b \cdot a$ ✓
- **Symmetric:** If $a \cdot d = b \cdot c$, then $c \cdot b = d \cdot a$ ✓
- **Transitive:** If $a \cdot d = b \cdot c$ and $c \cdot f = d \cdot e$, then $a \cdot f = b \cdot e$ ✓

Equivalence classes: Each equivalence class represents a rational number.

- $[\langle 1, 2 \rangle]_\sim = \{\langle 1, 2 \rangle, \langle 2, 4 \rangle, \langle 3, 6 \rangle, \langle -1, -2 \rangle, \ldots\}$ represents $\frac{1}{2}$
- $[\langle 3, 4 \rangle]_\sim = \{\langle 3, 4 \rangle, \langle 6, 8 \rangle, \langle -3, -4 \rangle, \ldots\}$ represents $\frac{3}{4}$

Mathematical significance: This construction shows how equivalence relations create new mathematical objects (rationals) from simpler ones (integers).

Quotient Sets

Definition 18: The *quotient set* of M by the equivalence relation R is the set of all equivalence classes:

$$M/_R = \{[x]_R \mid x \in M\}$$

Example: Consider $M = \{0, 1, 2, 3, 4, 5\}$ with the equivalence relation "congruent modulo 3":

$$x \equiv y \pmod 3 \quad \text{iff} \quad 3 \mid (x - y)$$

Equivalence classes:

- $[0]_\equiv = \{0, 3\} = \{x \in M \mid x \equiv 0 \pmod 3\}$
- $[1]_\equiv = \{1, 4\} = \{x \in M \mid x \equiv 1 \pmod 3\}$
- $[2]_\equiv = \{2, 5\} = \{x \in M \mid x \equiv 2 \pmod 3\}$

Quotient set:

$$M/_\equiv = \{\{0, 3\}, \{1, 4\}, \{2, 5\}\}$$

Note that $[3]_\equiv = [0]_\equiv = \{0, 3\}$, so we don't get a new equivalence class, since $M/_R$ is a *set*.

Quotient Sets [2]

Example: Let $M = \{\mathrm{a}, \mathrm{ab}, \mathrm{abc}, \mathrm{x}, \mathrm{xy}, \mathrm{z}\}$ with equivalence relation R defined by:

$$s_1 \; R \; s_2 \quad \text{iff} \quad |s_1| = |s_2|$$

where $|s|$ denotes the length of string s.

Equivalence classes by length:

- $[\mathrm{a}]_R = \{\mathrm{a}, \mathrm{x}, \mathrm{z}\}$ (strings of length 1)
- $[\mathrm{ab}]_R = \{\mathrm{ab}, \mathrm{xy}\}$ (strings of length 2)
- $[\mathrm{abc}]_R = \{\mathrm{abc}\}$ (strings of length 3)

Quotient set:

$$M/_R = \{\{\mathrm{a}, \mathrm{x}, \mathrm{z}\}, \{\mathrm{ab}, \mathrm{xy}\}, \{\mathrm{abc}\}\}$$

This partitions strings by their length, creating 3 equivalence classes.

Quotient Sets [3]

Example (Construction of rational numbers): Consider the set of ordered pairs $M = \mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$.

Define the equivalence relation $\sim$ by:

$$\langle a, b\rangle \sim \langle c, d\rangle \quad \text{iff} \quad a \cdot d = b \cdot c$$

Equivalence classes: Each represents a rational number in its "reduced form".

Representative	Equivalence Class	Rational Number
$\langle 1, 2\rangle$	$\{\langle 1, 2\rangle, \langle 2, 4\rangle, \langle 3, 6\rangle, \langle -1, -2\rangle, \langle -2, -4\rangle, \ldots\}$	$1/2$
$\langle 0, 1\rangle$	$\{\langle 0, 1\rangle, \langle 0, 2\rangle, \langle 0, -3\rangle, \langle 0, 7\rangle, \ldots\}$	0
$\langle 3, 4\rangle$	$\{\langle 3, 4\rangle, \langle 6, 8\rangle, \langle -3, -4\rangle, \langle 9, 12\rangle, \ldots\}$	$3/4$
$\langle -5, 3\rangle$	$\{\langle -5, 3\rangle, \langle 5, -3\rangle, \langle -10, 6\rangle, \langle 10, -6\rangle, \ldots\}$	$-5/3$

Quotient set: All equivalence classes together form the set of rational numbers:

$$\mathbb{Q} := (\mathbb{Z} \times (\mathbb{Z} \setminus \{0\}))/_\sim = \{[\langle a, b\rangle]_\sim \mid \langle a, b\rangle \in \mathbb{Z} \times (\mathbb{Z} \setminus \{0\})\}$$

Quotient Sets [4]

Interpretation: Each equivalence class $[\langle a, b\rangle]_\sim$ corresponds exactly to the rational number $\frac{a}{b}$.

- Different representations $\langle a, b\rangle$ and $\langle c, d\rangle$ belong to the same equivalence class iff they represent the same fraction: $\frac{a}{b} = \frac{c}{d}$
- The condition $a \cdot d = b \cdot c$ is the cross-multiplication test for fraction equality

Operations on the quotient set: We can define arithmetic operations on $\mathbb{Q}$ by:

$$[\langle a, b\rangle]_\sim + [\langle c, d\rangle]_\sim := [\langle ad + bc, bd\rangle]_\sim$$
$$[\langle a, b\rangle]_\sim \cdot [\langle c, d\rangle]_\sim := [\langle ac, bd\rangle]_\sim$$

These operations are *well-defined* because the result doesn't depend on the choice of representatives.

Mathematical significance:

- This construction shows that $\mathbb{Q}$ is literally *the* quotient set $\mathbb{Z} \times (\mathbb{Z} \setminus \{0\})/_\sim$
- It demonstrates how equivalence relations and quotient sets create new mathematical structures
- This is the rigorous foundation for rational number arithmetic taught in elementary school

Quotient Sets [5]

Example: Consider the set of all points in the plane $M = \mathbb{R}^2$ with the equivalence relation:

$$\langle x_1, y_1 \rangle \sim \langle x_2, y_2 \rangle \quad \text{iff} \quad x_1 + y_1 = x_2 + y_2$$

This relation groups points that lie on the same line of the form $x + y = c$ for some constant c.

Equivalence classes:

- $[\langle 0, 0 \rangle]_\sim = \{\langle x, y \rangle \mid x + y = 0\}$ (the line $x + y = 0$)
- $[\langle 1, 0 \rangle]_\sim = \{\langle x, y \rangle \mid x + y = 1\}$ (the line $x + y = 1$)
- $[\langle 0, 2 \rangle]_\sim = \{\langle x, y \rangle \mid x + y = 2\}$ (the line $x + y = 2$)

Quotient set:

$$\mathbb{R}^2/_\sim = \{L_c \mid c \in \mathbb{R}\}$$

where $L_c = \{\langle x, y \rangle \mid x + y = c\}$ is the line with equation $x + y = c$.

Each equivalence class is an entire line, and the quotient set consists of all such parallel lines.

Quotient Sets [6]

Example: Let $M = \{\text{apple}, \text{ant}, \text{banana}, \text{bee}, \text{cherry}, \text{cat}\}$ with equivalence relation:

$$w_1 \; R \; w_2 \quad \text{iff} \quad \text{first letter of } w_1 = \text{first letter of } w_2$$

Equivalence classes:

- $[\text{apple}]_R = \{\text{apple}, \text{ant}\}$ (words starting with 'a')
- $[\text{banana}]_R = \{\text{banana}, \text{bee}\}$ (words starting with 'b')
- $[\text{cherry}]_R = \{\text{cherry}, \text{cat}\}$ (words starting with 'c')

Quotient set:

$$M/_R = \{\{\text{apple}, \text{ant}\}, \{\text{banana}, \text{bee}\}, \{\text{cherry}, \text{cat}\}\}$$

This creates a dictionary-like grouping by first letter.

Quotient Sets [7]

Example (Trivial quotient sets):

Identity relation: If $R = I_M$ (identity relation), then each element forms its own equivalence class:

$$M/_{I_M} = \{\{x\} \mid x \in M\}$$

The quotient set has the same cardinality as the original set.

Universal relation: If $R = M \times M$ (universal relation), then all elements are equivalent:

$$M/_{M \times M} = \{M\}$$

The quotient set contains exactly one equivalence class — the entire set M.

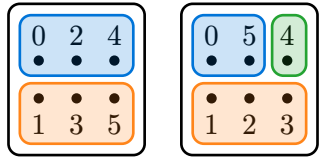
Set Partitions

Definition 19: A *partition* $\mathcal{P}$ of a set M is a family of non-empty, pairwise-disjoint subsets whose union is M:

- (Non-empty) $\forall B \in \mathcal{P}. (B \neq \emptyset)$
- (Disjoint) $\forall B_1, B_2 \in \mathcal{P}. (B_1 \neq B_2) \to (B_1 \cap B_2 = \emptyset)$
- (Cover) $\bigcup_{B \in \mathcal{P}} B = M$

Elements of $\mathcal{P}$ are *blocks* (or *cells*).

Example: For $M = \{0, 1, 2, 3, 4, 5\}$: $\{\{0, 2, 4\}, \{1, 3, 5\}\}$ and $\{\{0, 5\}, \{1, 2, 3\}, \{4\}\}$ are partitions.



Partitions and Equivalence Relations

Theorem 1 (Equivalences $\Leftrightarrow$ Partitions): Each equivalence relation R on M yields the partition $\mathcal{P}_R = \{[x]_R \mid x \in M\}$. Each partition $\mathcal{P}$ yields an equivalence $R_{\mathcal{P}}$ given by $\langle x, y\rangle \in R_{\mathcal{P}}$ iff x and y lie in the same block. These constructions invert one another.

Proof *(Sketch)*: Classes of an equivalence are non-empty, disjoint, and cover M. Conversely, "same block" relation is reflexive, symmetric, transitive. Composing the two constructions returns exactly the starting equivalence relation or partition (they are mutually inverse up to equality of sets of ordered pairs). $\square$

Composition of Relations

Composition of Relations

Definition 20: The *composition* of two relations $R \subseteq A \times B$ and $S \subseteq B \times C$ is defined as:

$$R \,;\, S = S \circ R = \{\langle a, c\rangle \mid \exists b \in B.\, (a \; R \; b) \wedge (b \; S \; c)\}$$

Example: Let $A = \{1, 2, 3\}$, $B = \{a, b, c, d\}$, $C = \{x, y\}$ with relations:

- $R = \{\langle 1, a\rangle, \langle 1, b\rangle, \langle 2, c\rangle, \langle 3, d\rangle\} \subseteq A \times B$
- $S = \{\langle a, x\rangle, \langle b, y\rangle, \langle c, x\rangle\} \subseteq B \times C$

To find $R \,;\, S$, we look for pairs $\langle i, z\rangle$ where there exists w such that $\langle i, w\rangle \in R$ and $\langle w, z\rangle \in S$:

From 1: can reach a and b via R

- a connects to x via $S \Rightarrow \langle 1, x\rangle$ is in the composition
- b connects to y via $S \Rightarrow \langle 1, y\rangle$ is in the composition

From 2: can reach c via R

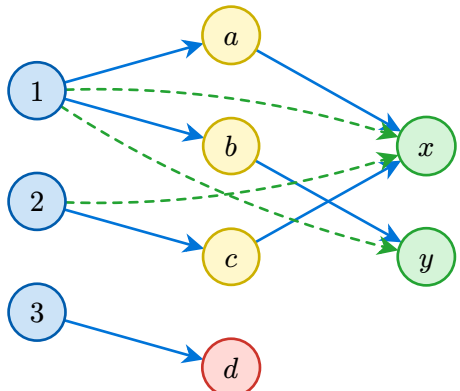
- c connects to x via $S \Rightarrow \langle 2, x\rangle$ is in the composition

From 3: can reach d via R

Composition of Relations [2]

- d has no outgoing connections in $S \Rightarrow$ no pairs from 3 in the composition

Therefore: $R \; ; \; S = \{\langle 1, x\rangle, \langle 1, y\rangle, \langle 2, x\rangle\}$



Edges legend:

- Solid: Relations R and S
- Dashed: Composition $R \; ; \; S$
- Red: Dead end (no outgoing path)

Examples of Composition

Example (Composition in a social network): Consider three sets:

- People $= \{\text{Alice}, \text{Bob}, \text{Carol}\}$
- Skills $= \{\text{Python}, \text{Design}, \text{Management}\}$
- Projects $= \{\text{WebApp}, \text{Mobile}, \text{Analytics}\}$

Define relations:

- HasSkill $= \{\langle \text{Alice}, \text{Python}\rangle, \langle \text{Alice}, \text{Design}\rangle, \langle \text{Bob}, \text{Python}\rangle, \langle \text{Carol}, \text{Management}\rangle\}$
- RequiresSkill $= \{\langle \text{Python}, \text{WebApp}\rangle, \langle \text{Python}, \text{Analytics}\rangle, \langle \text{Design}, \text{WebApp}\rangle, \langle \text{Management}, \text{Mobile}\rangle\}$

The composition HasSkill ; RequiresSkill gives us CanWorkOn:

- Alice can work on WebApp (via Python AND Design)
- Alice can work on Analytics (via Python)
- Bob can work on WebApp (via Python)
- Bob can work on Analytics (via Python)
- Carol can work on Mobile (via Management)

So: CanWorkOn $=$
$\{\langle \text{Alice}, \text{WebApp}\rangle, \langle \text{Alice}, \text{Analytics}\rangle, \langle \text{Bob}, \text{WebApp}\rangle, \langle \text{Bob}, \text{Analytics}\rangle, \langle \text{Carol}, \text{Mobile}\rangle\}$

Examples of Composition [2]

Example (Matrix composition): Relations can be composed using Boolean matrix multiplication.

Given $R \subseteq \{1, 2\} \times \{a, b\}$ and $S \subseteq \{a, b\} \times \{x, y\}$:

$$[\![R]\!] = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \quad [\![S]\!] = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

The composition $[\![R \mathbin{;} S]\!] = [\![R]\!] \odot [\![S]\!]$ using Boolean matrix multiplication:

$$[\![R \mathbin{;} S]\!] = \begin{bmatrix} (1 \wedge 1) \vee (0 \wedge 0) & (1 \wedge 1) \vee (0 \wedge 1) \\ (1 \wedge 1) \vee (1 \wedge 0) & (1 \wedge 1) \vee (1 \wedge 1) \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

This means $R \mathbin{;} S = \{\langle 1, x\rangle, \langle 1, y\rangle, \langle 2, x\rangle, \langle 2, y\rangle\}$ (the universal relation).

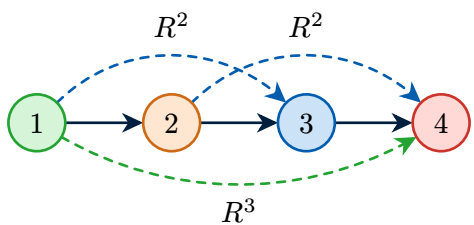
Powers of Relations

Definition 21: For a homogeneous relation $R \subseteq M^2$, we define *powers* of R:

- $R^0 = I_M$ (identity relation)
- $R^1 = R$
- $R^{n+1} = R^n \circ R$ for $n \geq 1$

Example: Let $M = \{1, 2, 3, 4\}$ and $R = \{\langle 1, 2\rangle, \langle 2, 3\rangle, \langle 3, 4\rangle\}$ (successor relation).

- $R^1 = \{\langle 1, 2\rangle, \langle 2, 3\rangle, \langle 3, 4\rangle\}$
- $R^2 = \{\langle 1, 3\rangle, \langle 2, 4\rangle\}$ (two steps)
- $R^3 = \{\langle 1, 4\rangle\}$ (three steps)
- $R^4 = \emptyset$ (no four-step paths)

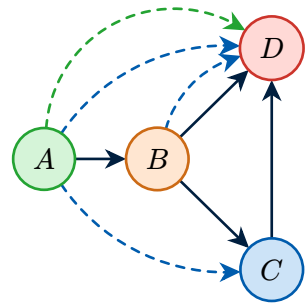


Paths in Graphs

Example (Path composition in a graph): Consider a directed graph with vertices $\{A, B, C, D\}$ and relation R representing direct edges: $R = \{\langle A, B\rangle, \langle B, C\rangle, \langle B, D\rangle, \langle C, D\rangle\}$

Powers of R represent paths of different lengths:

- $R^1 = R$ (direct connections)
- $R^2 = R \circ R$ (2-step paths):
  - $\langle A, C\rangle$: path $A \to B \to C$
  - $\langle A, D\rangle$: path $A \to B \to D$
  - $\langle B, D\rangle$: path $B \to C \to D$
  - So $R^2 = \{\langle A, C\rangle, \langle A, D\rangle, \langle B, D\rangle\}$.
- $R^3 = R^2 \circ R$ (3-step paths):
  - $\langle A, D\rangle$: path $A \to B \to C \to D$
  - No more 3-step paths!
  - So $R^3 = \{\langle A, D\rangle\}$.
- $R^4 = \emptyset$ (no 4-step paths)



Associativity of Composition

Theorem 2: Composition of relations is associative: $(R \mathbin{;} S) \mathbin{;} T = R \mathbin{;} (S \mathbin{;} T)$.

Proof: Let $R \subseteq A \times B$, $S \subseteq B \times C$, and $T \subseteq C \times D$ be three relations.

($\subseteq$): Let $\langle a, d\rangle \in (R \mathbin{;} S) \mathbin{;} T$.

- By definition of composition: $\exists c \in C. (\langle a, c\rangle \in R \mathbin{;} S) \wedge (\langle c, d\rangle \in T)$.
- Since $\langle a, c\rangle \in R \mathbin{;} S$, we have: $\exists b \in B. (\langle a, b\rangle \in R) \wedge (\langle b, c\rangle \in S)$.
- From $\langle b, c\rangle \in S$ and $\langle c, d\rangle \in T$, we have: $\langle b, d\rangle \in S \mathbin{;} T$.
- From $\langle a, b\rangle \in R$ and $\langle b, d\rangle \in S \mathbin{;} T$, we have: $\langle a, d\rangle \in R \mathbin{;} (S \mathbin{;} T)$.

($\supseteq$): Let $\langle a, d\rangle \in R \mathbin{;} (S \mathbin{;} T)$.

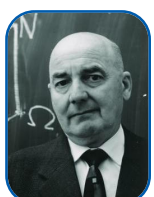
- By definition of composition: $\exists b \in B. (\langle a, b\rangle \in R) \wedge (\langle b, d\rangle \in S \mathbin{;} T)$.
- Since $\langle b, d\rangle \in S \mathbin{;} T$, we have: $\exists c \in C. (\langle b, c\rangle \in S) \wedge (\langle c, d\rangle \in T)$.
- From $\langle a, b\rangle \in R$ and $\langle b, c\rangle \in S$, we have: $\langle a, c\rangle \in R \mathbin{;} S$.
- From $\langle a, c\rangle \in R \mathbin{;} S$ and $\langle c, d\rangle \in T$, we have: $\langle a, d\rangle \in (R \mathbin{;} S) \mathbin{;} T$.

Therefore, $(R \mathbin{;} S) \mathbin{;} T = R \mathbin{;} (S \mathbin{;} T)$. $\square$

Order Theory

"Order is heaven's first law."

— Alexander Pope



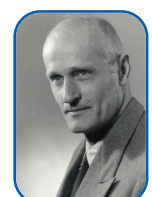
Helmut Hasse

Felix Hausdorff

Henri Poincaré

Robert Floyd

Stephen Warshall



Stephen Kleene

Orders

Definition 22: A relation $R \subseteq M^2$ is called a *preorder* if it is reflexive and transitive.

Definition 23: A *partial order* is a relation $R \subseteq M^2$ that is reflexive, antisymmetric, and transitive.

Definition 24: A *linear order* is a partial order $R \subseteq M^2$ where every pair of elements is *comparable*:

$$\forall x, y \in M. (x \; R \; y \lor y \; R \; x)$$

Note: The above condition is called *strong connectivity*. The similar property with the $x \neq y$ condition is called *semi-connectivity*. The corresponding relations are called *connex* and *semi-connex*.

Note: Hereinafter, we mainly study the "non-strict" orders (*e.g.*, $\leq$, $\subseteq$), which require the *reflexivity*. The "strict" orders (*e.g.*, $<$, $\subset$) require *irreflexivity* instead, and are defined similarly.

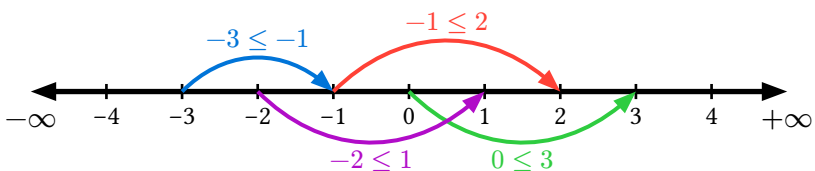
Note: Strict linear orders require *semi-connectivity* instead, since connectivity alone implies reflexivity!

Examples of Orders

Example: The relation $\leq$ on real numbers $\mathbb{R}$ is a *total order*:

- **Reflexive:** $x \leq x$ for all $x \in \mathbb{R}$ ✓
- **Antisymmetric:** If $x \leq y$ and $y \leq x$, then $x = y$ ✓
- **Transitive:** If $x \leq y$ and $y \leq z$, then $x \leq z$ ✓
- **Connected:** For any $x, y \in \mathbb{R}$, either $x \leq y$ or $y \leq x$ ✓

This is the most familiar example of an order relation. Similarly, $\mathbb{N}$, $\mathbb{Z}$, and $\mathbb{Q}$ with $\leq$ are all total orders.



Linear order $\leq$ on real numbers $\mathbb{R}$

Examples of Orders [2]

Example: The *subset relation* $\subseteq$ on the power set $\mathcal{P}(A)$ is a **partial order**.

Verification:

- **Reflexive:** $X \subseteq X$ for all $X \subseteq A$ ✓
  - Every set is a subset of itself by definition
- **Antisymmetric:** If $X \subseteq Y$ and $Y \subseteq X$, then $X = Y$ ✓
  - If every element of X is in Y, and every element of Y is in X, then X and Y have the same elements
- **Transitive:** If $X \subseteq Y$ and $Y \subseteq Z$, then $X \subseteq Z$ ✓
  - If every element of X is in Y, and every element of Y is in Z, then every element of X is in Z

For $A = \{1, 2\}$, we have $\mathcal{P}(A) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$ with:

- $\emptyset \subseteq \{1\} \subseteq \{1, 2\}$ (this is a chain)
- $\emptyset \subseteq \{2\} \subseteq \{1, 2\}$ (another chain)
- But $\{1\}$ and $\{2\}$ are *incomparable* (neither is a subset of the other)

This is *not* a total order because not all pairs are comparable.

Examples of Orders [3]

Example: Consider the *no longer than* relation $\preccurlyeq$ on binary strings $\mathbb{B}^*$:

$$x \preccurlyeq y \quad \text{iff} \quad \text{len}(x) \leq \text{len}(y)$$

Verification:

- **Reflexive:** $x \preccurlyeq x$ for all $x \in \mathbb{B}^*$ ✓
  - $\text{len}(x) \leq \text{len}(x)$ is always true
- **Transitive:** If $x \preccurlyeq y$ and $y \preccurlyeq z$, then $x \preccurlyeq z$ ✓
  - If $\text{len}(x) \leq \text{len}(y)$ and $\text{len}(y) \leq \text{len}(z)$, then $\text{len}(x) \leq \text{len}(z)$ by transitivity of $\leq$
- **Antisymmetric:** ✗
  - Counter-example: $01 \preccurlyeq 10$ and $10 \preccurlyeq 01$ (since both have length 2), but $01 \neq 10$
- **Connected:** For any $x, y \in \mathbb{B}^*$, either $x \preccurlyeq y$ or $y \preccurlyeq x$ ✓
  - Either $\text{len}(x) \leq \text{len}(y)$ or $\text{len}(y) \leq \text{len}(x)$ (or both)

This is a *preorder* (reflexive and transitive), and even connected, but *not a partial order* due to lack of antisymmetry. Different strings of the same length are all "equivalent" under this relation, but they're not actually equal.

Examples of Orders [4]

Example: *Divisibility* $\mid$ on positive integers $\mathbb{N}^+$ is a **partial order**.

Verification:

- **Reflexive:** $n \mid n$ for all $n \in \mathbb{N}^+$ (every number divides itself) ✓
- **Antisymmetric:** If $a \mid b$ and $b \mid a$, then $a = b$ ✓
  - If a divides b, then $b = ak$ for some positive integer k
  - If b divides a, then $a = b\ell$ for some positive integer ℓ
  - Substituting: $b = ak = (b\ell)k = bk\ell$, so $k\ell = 1$
  - Since $k, \ell \in \mathbb{N}^+$, we must have $k = \ell = 1$, hence $a = b$
- **Transitive:** If $a \mid b$ and $b \mid c$, then $a \mid c$ ✓
  - If $a \mid b$, then $b = ak$ for some integer k
  - If $b \mid c$, then $c = b\ell$ for some integer ℓ
  - Therefore: $c = b\ell = (ak)\ell = a(k\ell)$, so $a \mid c$
- **Connected:** ✗
  - Counter-example: 2 does not divide 3, and 3 does not divide 2

This is a *partial order* but not a total order because some pairs are incomparable.

Examples of Orders [5]

Example: *Lexicographic order* on strings A^n (like dictionary order) is a **total order**.

Verification:

- **Reflexive:** $s \preceq s$ for all strings s ✓
  - A string is lexicographically equal to itself
- **Antisymmetric:** If $s \preceq t$ and $t \preceq s$, then $s = t$ ✓
  - If s comes before or equals t AND t comes before or equals s, then $s = t$
- **Transitive:** If $s \preceq t$ and $t \preceq u$, then $s \preceq u$ ✓
  - Lexicographic comparison preserves transitivity through character-by-character comparison
- **Connected:** For any strings s, t, either $s \preceq t$ or $t \preceq s$ ✓
  - We can always compare strings lexicographically by comparing character by character

For binary strings of length 2: $00 \prec 01 \prec 10 \prec 11$

Key property: Every pair of strings is *comparable*, making this a *total order*.

Partially Ordered Sets

Definition 25: A *partially ordered set* (or *poset*) $\langle S, \leq \rangle$ is a set S equipped with a partial order $\leq$.

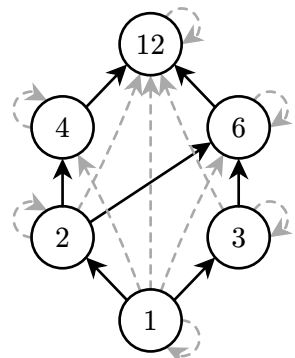
Example: Consider the poset $\langle D, | \rangle$ where $D = \{1, 2, 3, 4, 6, 12\}$ and $|$ is divisibility.

Order relation properties:

- Reflexive: $n \mid n$ for all $n \in D$
- Antisymmetric: If $a \mid b$ and $b \mid a$, then $a = b$
- Transitive: If $a \mid b$ and $b \mid c$, then $a \mid c$

Special elements:

- Least element: 1 (divides all others)
- Greatest element: 12 (divisible by all others)
- Minimal elements: just 1
- Maximal elements: just 12



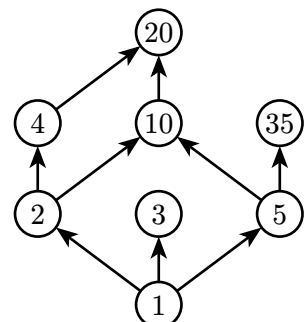
Note: This poset forms a *lattice* since every pair has a supremum (LCM) and infimum (GCD).

Hasse Diagrams

Definition 26: A *Hasse diagram* is a visual representation of a poset where:

- Each element is represented as a vertex.
- If $x < y$ and there is no z with $x < z < y$, draw an edge from x to y.
- Elements are arranged vertically by the order relation.
- Transitive connections are omitted (implied by transitivity of the partial order).

Example: For $D = \{1, 2, 3, 4, 5, 10, 20, 35\}$ with divisibility $|$.

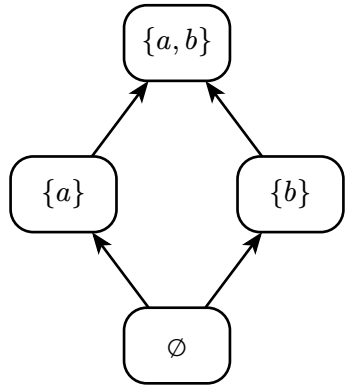


Reading the diagram:

- 1 divides everything (least element at bottom)
- Multiple *maximal* elements: 3, 20, 35 (no single greatest element)
- Some *chains* (not necessarily maximal, can skip elements):
  - Chain: $1 \mid 2 \mid 4$ (powers of 2)
  - Chain: $1 \mid 10 \mid 20$ (multiples of 10)
  - Chain: $5 \mid 35$ (multiples of 5)
- Primes 2, 3, 5 are *incomparable* to each other

Subset Poset

Example: For $\mathcal{P}(\{a, b\}) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$ with inclusion $\subseteq$:



Note: This is the classic "diamond" shape characteristic of Boolean algebras.

Note: $\langle \mathcal{P}(A), \subseteq \rangle$ is also called the *Boolean lattice*.

Covering Relation

Definition 27: In a poset $\langle S, \leq \rangle$, an element $y \in S$ *covers* $x \in S$, denoted $x \lessdot y$, if there is no other element in between them:

$$x \lessdot y \quad \text{iff} \quad (x < y) \wedge \nexists z \in S. (x < z < y)$$

Note: "$<$" denotes the *induced strict order*:

$$x < y \quad \text{iff} \quad (x \leq y) \wedge (x \neq y)$$

Note: Hasse diagram is just a graph of a covering relation!

Maximal and Minimal Elements

Definition 28: An element $m \in S$ is called a *maximal element* of a poset $\langle S, \leq \rangle$ if it is not less than any other element, *i.e.*, there is no even greater element.

$$\forall x \neq m. \neg(m \leq x) \quad \Longleftrightarrow \quad \nexists x \neq m. (m \leq x)$$

Equivelently, $\forall x \in S. (m \leq x) \rightarrow (m = x)$

Definition 29: An element $m \in S$ is called a *minimal element* of a poset $\langle S, \leq \rangle$ if it is not greater than any other element, *i.e.*, there is no even smaller element.

$$\forall x \neq m. \neg(x \leq m) \quad \Longleftrightarrow \quad \nexists x \neq m. (x \leq m)$$

Equivelently, $\forall x \in S. (x \leq m) \rightarrow (x = m)$

Note: There may be multiple maximal (or minimal) elements.

Example of Maximal and Minimal Elements

Example: Consider the divisibility poset on $S = \{2, 3, 4, 6, 8, 12\}$:

Maximal elements: 8 and 12

- 8 divides nothing else in S except itself
- 12 divides nothing else in S except itself

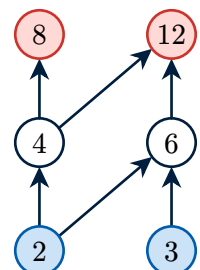
Minimal elements: 2 and 3

- Nothing in S properly divides 2 (since $1 \notin S$)
- Nothing in S properly divides 3

Note: This poset has no greatest or least element, but multiple minimal/maximal elements.

Hasse diagram:

- Maximal elements (8, 12) — they divide nothing else in S
- Minimal elements (2, 3) — nothing in S divides them
- Two separate chains: $2 \mid 4 \mid 8$ and $3 \mid 6 \mid 12$



Example of Maximal and Minimal Elements [2]

Example: Let $S = \{\text{ab}, \text{abc}, \text{abd}, \text{ac}, \text{b}, \text{bc}\}$ ordered by the *prefix* relation $\lessdot$:

$$x \lessdot y \text{ iff } x \text{ is a prefix of } y$$

Maximal elements: abc, abd, <u>ac</u>, bc

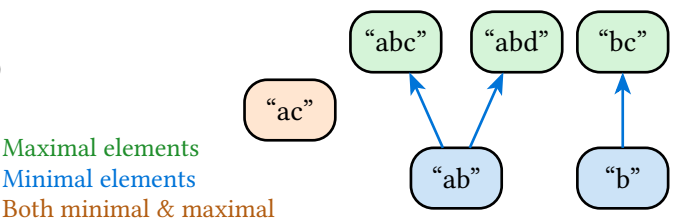
- These strings are not prefixes of any other string in S

Minimal elements: ab, <u>ac</u>, b

- These strings have no other string in S that is a proper prefix of them

The Hasse diagram shows three separate "trees" rooted at the minimal elements, with:

- $\text{ab} \lessdot \text{abc}$ and $\text{ab} \lessdot \text{abd}$
- $\text{b} \lessdot \text{bc}$
- ac stands alone (no other string extends it in S)



Greatest and Least Elements

Definition 30: The *greatest element* of a poset $\langle S, \leq \rangle$ is an element $g \in S$ that is greater than or equal to every other element in S, *i.e.*, for all $x \in S$, $x \leq g$.

Definition 31: A *least element* (bottom) b satisfies $b \leq x$ for all $x \in S$.

Note: Greatest (top) and least (bottom) elements are *unique* when they exist.

Examples:

- $\langle \mathcal{P}(A), \subseteq \rangle$: least $\emptyset$ (contained in every set), greatest A (contains every subset).
- $\langle \mathbb{N}^+, | \rangle$: least 1 (divides every positive integer), no greatest element (no integer is divisible by all others).
- $\langle \mathbb{Z}, \leq \rangle$: no least or greatest element (integers extend infinitely in both directions).
- $\langle \{1, 2, 3, 4, 5, 6\}, | \rangle$: least 1, no greatest element, maximal elements are 4, 5, 6 (prime powers and primes that don't divide anything else in the given set).

Converse Orders

Definition 32: The *dual* (or *converse*) of a poset $\langle S, \leq \rangle$ is the poset $\langle S, \geq \rangle$ where $x \geq y$ iff $y \leq x$.

Example: Consider the $\mathbb{N}^+$ ordered naturally.

- For $\leq$ order:
  - The *$\leq$-least* element is 1, since it is *$\leq$-smaller* than all others.
  - 1 is also $\leq$-minimal, since there are no other element which is $\leq$-smaller than 1.
  - There are *no $\leq$-maximal* elements, since the set is unbounded above.
  - On the Hasse diagram, 1 is at the bottom, and the diagram extends infinitely *upwards*.
- For $\geq$ order, minimal and maximal elements "flip":
  - There are *no $\geq$-minimal* elements. A $\geq$-minimal element would be an element m such that there is no other ($\geq$-smaller) element $n \neq m$ with $n \geq m$. However, for any $m \in \mathbb{N}^+$, there exists $n = m + 1$ which is $n \geq m$, so no such $\geq$-minimal element exists.
  - The *$\geq$-greatest* element is 1, since all elements are *$\geq$-smaller* than it.
  - On the Hasse diagram, 1 is at the top, and the diagram extends infinitely *downwards*.

Notes on Converse Orders

Note:

- Maximal elements in $\langle S, \leq \rangle$ become minimal in $\langle S, \geq \rangle$ and vice versa.
- Greatest element in $\langle S, \leq \rangle$ becomes least in $\langle S, \geq \rangle$ and vice versa.
- Chains and antichains remain the same in both orders.
- The Hasse diagram is flipped vertically when taking the dual order.
  - For $\langle \mathbb{N}^+, \leq \rangle$:
    - The *least* element 1 is at the bottom.
    - The diagram of $\langle \mathbb{N}^+, \leq \rangle$ extends infinitely *upwards*.
  - In the dual $\langle \mathbb{N}^+, \geq \rangle$:
    - 1 becomes the *greatest* element at the top.
    - The diagram of $\langle \mathbb{N}^+, \geq \rangle$ extends infinitely *downwards*.

Chains and Antichains

Definition 33: In a partially ordered set $\langle M, \leq \rangle$:

- A *chain* is a subset $C \subseteq M$ where every two elements are comparable. Formally:

$$\forall x, y \in C. \, (x \leq y \text{ or } y \leq x)$$

- An *antichain* is a subset $A \subseteq M$ where no two distinct elements are comparable. Formally:

$$\forall x, y \in A. \, (x \neq y) \rightarrow (x \not\leq y \text{ and } y \not\leq x)$$

Note:

- Chains correspond to arbitrary paths or sub-sequences in the Hasse diagram.
- A *maximal chain* is a chain that cannot be extended by including any other elements from M.
- A *maximum chain* is a chain of the largest possible size in M.
- A chain is a *totally ordered subset* of the poset.
- An antichain consists of *pairwise incomparable elements*.
- Any singleton set is both a chain and an antichain.

Examples of Chains and Antichains

Example: Consider the divisibility poset $\langle D, | \rangle$ where $D = \{1, 2, 3, 4, 5, 6, 10, 20, 35\}$.

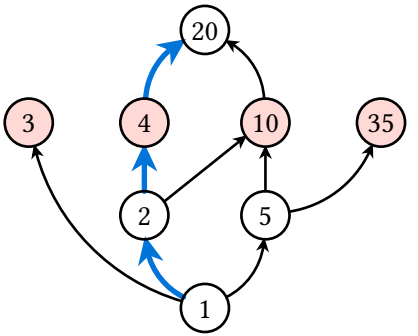
Chains: (totally ordered subsets)

- Maximal chain: $\{1, 2, 4, 20\}$ — longest path
- Not maximal: $\{5, 10\}$
- Can *skip* elements: $\{1, 5, 20\}$

Maximal elements: 3, 20, 35

Antichains: (pairwise incomparable elements)

- Maximal antichain: $\{3, 4, 10, 35\}$
- Incomparable on the same level: $\{2, 5\}$
- Incomparable from different levels: $\{3, 2, 35\}$



Dilworth's theorem: Maximum antichain size (4) = minimum chains needed to cover (4).

Examples of Chains and Antichains [2]

Example: In a Git repository, commits form a poset under the "ancestor" relation:

- **Chain:** A sequence of commits on a single branch (linear history).
- **Antichain:** Commits on different branches that have diverged (no ancestry relation).

Practical insight: Merge commits combine multiple antichains back into a single chain.

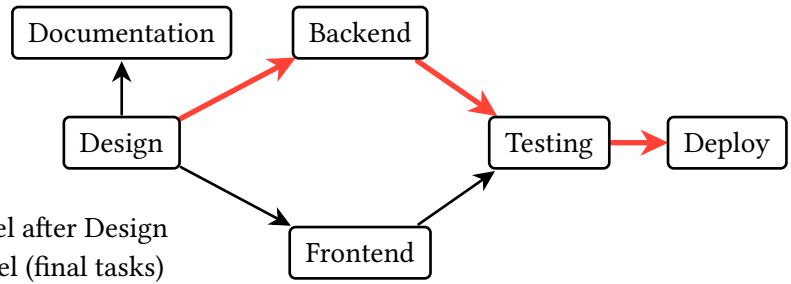
Chains and Antichains in Scheduling

Example: In project management, tasks form a scheduling poset under the "*prerequisite*" relation.

Consider web development tasks: Design, Backend, Frontend, Testing, Deploy, Documentation.

Dependencies:

- Design $\prec$ Back, Front
- Back, Front $\prec$ Test
- Test $\prec$ Deploy $\prec$ Doc



Antichain analysis:

- {Back, Front} can run in parallel after Design
- {Deploy, Doc} can run in parallel (final tasks)

Critical path: Design $\rightarrow$ Back $\rightarrow$ Test $\rightarrow$ Deploy (length 3 max chain)

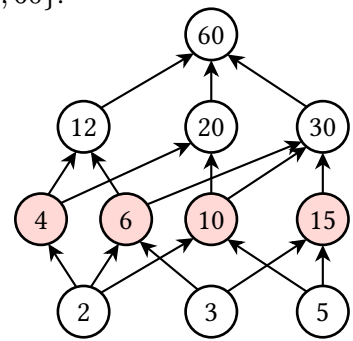
Practical insights:

- Chains = sequential dependencies (critical path)
- Antichains = tasks for parallel execution (resource allocation)
- Project duration = length of longest chain

Dilworth's Theorem

Theorem 3 (Dilworth): In any finite partially ordered set, the maximum size of an antichain equals the minimum number of chains needed to cover the entire set.

Example: Consider the divisibility poset on $P = \{2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60\}$.



Maximum antichain: $\{4, 6, 10, 15\}$ (size 4)

- These elements are pairwise *incomparable* (none divides another).

Minimum chain decomposition: We need exactly 4 chains that *cover* P:

- Chain 1: $2 \mid 4 \mid 12 \mid 60$
- Chain 2: $3 \mid 6 \mid 30$
- Chain 3: $5 \mid 10 \mid 20$
- Chain 4: 15 (singleton chain)

Each element appears in exactly one chain, forming a *partition* of P.

Dilworth's theorem: Maximum *antichain* size (4) = Minimum number of *disjoint chains* (4). ✓

Proof of Dilworth's Theorem

Proof: Let $\langle P, \leq \rangle$ be a finite poset. Let α denote the maximum size of an antichain in P, and let β denote the minimum number of chains needed to cover P. We prove $\alpha = \beta$ by showing $\alpha \leq \beta$ and $\alpha \geq \beta$.

Easy part ($\alpha \leq \beta$): Suppose P can be partitioned into k chains $C_1, \ldots, C_k$. Let A be any antichain in P. Since elements in an antichain are pairwise incomparable, each chain contains at most one element of A. Therefore $|A| \leq k$. Taking the maximum over all antichains gives $\alpha \leq k$. Since this holds for any chain partition, we have $\alpha \leq \beta$. ■

Proof of Dilworth's Theorem [2]

Hard part ($\alpha \geq \beta$): Let $A \subseteq P$ be a maximal antichain of size α.

We construct a chain partition of P of size α as follows:

- Initialize $\mathcal{C} := \emptyset$.
- While $P \neq \emptyset$:
  1. Choose a maximal element $x \in P$ (no element above x in the remaining poset).
  2. Build a chain C ending at x:
     - Start with $C := \{x\}$.
     - Repeatedly add a maximal *predecessor* element $y \in P \setminus C$ such that $y <$ current bottom of C.
  3. Add C to $\mathcal{C}$ and remove all elements of C from P.

By construction:

- Each $C \in \mathcal{C}$ is a chain (elements are added only below the current bottom).
- Chains in $\mathcal{C}$ cover all elements of P.
- Each chain contains exactly one element of the maximal antichain A, so $|\mathcal{C}| = \alpha$.

Therefore P can be covered by α chains, giving $\beta \leq \alpha$. $\square$

Summary: Orders

Orders provide structured ways to compare and rank elements:

- *Preorders:* Basic comparison (reflexive, transitive)
- *Partial orders:* Add antisymmetry for unique comparisons
- *Total orders:* Every pair of elements is comparable

Visualization: Hasse diagrams clearly show structure and hierarchy by omitting redundant transitive edges, revealing *chains* (ordered sequences) and *antichains* (incomparable elements).

Applications: Task scheduling • Git version control • Database indexing • Type hierarchies • Boolean algebra • Concurrent systems • Build systems (dependency resolution) • Social networks • File system permissions • Web page ranking • Package managers • Distributed systems

Functions

"A function is a machine which converts a certain class of inputs into a certain class of outputs."

— Norbert Wiener

Leonhard Euler

Augustin-Louis Cauchy

Karl Weierstrass

Joseph-Louis Lagrange

George Pólya

Norbert Wiener

Definition of a Function

Definition 34: A *function* f from a set A to a set B, denoted $f : A \to B$, is a special kind of relation $f \subseteq A \times B$ where every element of A is paired with *exactly one* element of B.

This "exactly one" requirement breaks down into two conditions:

1. *Functional property* (*right-unique* or *well-defined*): Each input has *at most one* output. No input can map to multiple different outputs.

$$\forall a \in A. \forall b_1, b_2 \in B. (f(a) = b_1) \land (f(a) = b_2) \to (b_1 = b_2)$$

2. *Total property* (*left-total* or *defined everywhere*): Each input has *at least one* output. Every element in the domain must map to something.

$$\forall a \in A. \exists b \in B. f(a) = b$$

Definition of a Function [2]

Definition 35: A relation that satisfies only the *functional* property (but not necessarily total) is called a *partial function*, denoted $f : A \hookrightarrow B$. It may be undefined for some inputs.

Definition 36: A relation that satisfies *both* properties is called a *total function*, denoted $f : A \to B$. It is defined for every input in its domain.

Note: In most contexts, when we say "function" we mean *total function*. Partial functions are explicitly noted when needed, especially in computability theory and programming language semantics.

Examples of Functions

Example (Partial functions):

- **Division**: $f : \mathbb{R} \times \mathbb{R} \hookrightarrow \mathbb{R}$ defined by $f(x, y) = \frac{x}{y}$ is partial because $f(x, 0)$ is undefined.
- **Square root on integers**: $g : \mathbb{Z} \hookrightarrow \mathbb{R}$ defined by $g(n) = \sqrt{n}$ is partial because negative integers have no real square root.
- **Array access**: $\text{arr} : \mathbb{N} \hookrightarrow T$ where $\text{arr}(i)$ returns the element at index i, but is undefined for out-of-bounds indices.
- **Head of list**: $\text{head} : \text{List}[T] \hookrightarrow T$ returns the first element, but is undefined for empty lists.

Example (Total functions):

- **Absolute value**: $|\cdot| : \mathbb{R} \to \mathbb{R}$ is total — defined for all real numbers.
- **Successor**: $S : \mathbb{N} \to \mathbb{N}$ defined by $S(n) = n + 1$ is total — defined for all natural numbers.
- **Constant function**: $f : A \to B$ defined by $f(x) = b_0$ for some fixed $b_0 \in B$ is always total.

Domain, Codomain, Range

Definition 37: For a function $f : A \to B$:

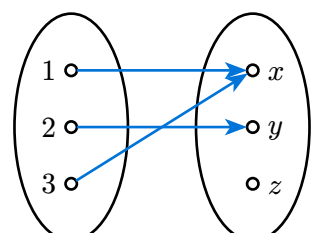
- The set A is called the *domain* of f, denoted $\mathrm{Dom}(f)$.
- The set B is called the *codomain* of f, denoted $\mathrm{Cod}(f)$.
- The *range* (or *image*) of f is the set of all values that f actually takes:

$$\mathrm{Range}(f) = \{b \in B \mid \exists a \in A.\, f(a) = b\} = \{f(a) \mid a \in A\}$$

Note: $\mathrm{Range}(f) \subseteq \mathrm{Cod}(f)$

Example: Let $A = \{1, 2, 3\}$ and $B = \{x, y, z\}$, and define $f = \{\langle 1, x\rangle, \langle 2, y\rangle, \langle 3, x\rangle\}$.

- $f : A \to B$ is a function from A to B
- $\mathrm{Dom}(f) = A = \{1, 2, 3\}$ ("from")
- $\mathrm{Cod}(f) = B = \{x, y, z\}$ ("to")
- $\mathrm{Range}(f) = \{x, y\} \subset B$ (note that z is not in the range)
- We have $f(1) = x$, $f(2) = y$, $f(3) = x$



Examples of Domain, Codomain, Range

Example: Consider the *squaring* function $g : \mathbb{Z} \to \mathbb{Z}$ defined by $g(n) = n^2$.

- $\mathrm{Dom}(g) = \mathrm{Cod}(g) = \mathbb{Z}$ (all integers)
- $\mathrm{Range}(g) = \{0, 1, 4, 9, 16, \ldots\} = \{n^2 \mid n \in \mathbb{N}\}$ (non-negative perfect squares)

Note that the range of g is a proper subset of the codomain: $\mathrm{Range}(g) \subset \mathrm{Cod}(g)$, since $-1 \notin \mathrm{Range}(g)$.

Example: The *absolute value* function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = |x|$:

- $\mathrm{Dom}(f) = \mathrm{Cod}(f) = \mathbb{R}$ (all real numbers)
- $\mathrm{Range}(f) = [0, \infty) = \{y \in \mathbb{R} \mid y \geq 0\}$ (non-negative reals)

Example: The *exponential* function $\exp : \mathbb{R} \to (0, \infty)$ defined by $\exp(x) = e^x$:

- $\mathrm{Dom}(\exp) = \mathbb{R}$ (all real numbers)
- $\mathrm{Cod}(\exp) = (0, \infty)$ (positive real numbers)
- $\mathrm{Range}(\exp) = (0, \infty)$ (same as codomain — this function is *surjective*!)

Note the careful choice of codomain: if we used $\exp : \mathbb{R} \to \mathbb{R}$, the range would still be $(0, \infty)$, making it not surjective.

Image and Preimage of Sets

Definition 38: Let $f : A \to B$ be a function and let $S \subseteq A$. The *image of S under f* is the set:

$$f(S) = \{f(s) \mid s \in S\}$$

Note that $f(S) \subseteq B$. The range of f is $f(A)$.

Definition 39: Let $f : A \to B$ be a function and let $T \subseteq B$. The *preimage of T under f* (or *inverse image of T*) is the set of all elements in the domain that map into T:

$$f^{-1}(T) = \{a \in A \mid f(a) \in T\}$$

Note: The notation $f^{-1}(S)$ is used even if the inverse function f^{-1} does not exist (*i.e.*, if f is not bijective). It always refers to the set of domain elements that map into S.

Examples of Image and Preimage

Example: Let $f : \mathbb{Z} \to \mathbb{Z}$ be $f(x) = x^2$.

- Let $S = \{-2, -1, 0, 1, 2\}$.
  - Then $f(S) = \{f(-2), f(-1), f(0), f(1), f(2)\} = \{4, 1, 0, 1, 4\} = \{0, 1, 4\}$.
- Let $T_1 = \{1, 9\}$.
  - The preimage is $f^{-1}(T_1) = \{x \in \mathbb{Z} \mid x^2 \in \{1, 9\}\} = \{-3, -1, 1, 3\}$.
- Let $T_2 = \{2, 3\}$.
  - The preimage is $f^{-1}(T_2) = \{x \in \mathbb{Z} \mid x^2 \in \{2, 3\}\} = \emptyset$.

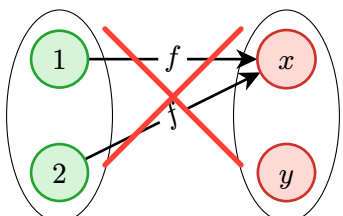
Injective Functions

Definition 40: A function $f : A \to B$ is *injective* (*left-unique* or *one-to-one*[3]) if distinct elements in the domain map to distinct elements in the codomain. Formally:

$$\forall a_1, a_2 \in A. \, (f(a_1) = f(a_2)) \to (a_1 = a_2)$$

Equivalent formulation:

$$\forall a_1, a_2 \in A. \, (a_1 \neq a_2) \to (f(a_1) \neq f(a_2))$$



[3]Do not confuse "one-to-one" with "one-to-one correspondence", which refers to a *bijection* — another concept!

Examples of Injective Functions

Key insight: An injective function never "collapses" different inputs to the same output. Each output has at most one pre-image.

Example: $f : \mathbb{N} \to \mathbb{N}$ defined by $f(n) = 2n$ is injective.

- If $f(n_1) = f(n_2)$, then $2n_1 = 2n_2$, so $n_1 = n_2$. ✓

Example: $g : \mathbb{R} \to \mathbb{R}$ defined by $g(x) = x^3$ is injective.

- The cube function is strictly increasing, so distinct inputs give distinct outputs. ✓

Example (Computer science applications):

- **Student ID assignment:** Each student gets a unique ID number.
- **Hash functions (perfect):** Different data should hash to different values (to avoid collisions).
- **Primary keys in databases:** Each record must have a unique identifier.
- **Memory addresses:** Each memory location has a unique address.

Examples of Non-Injective Functions

Example (Counter-examples (not injective functions)):

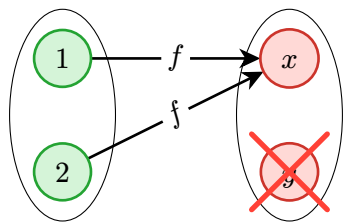
- $g : \mathbb{Z} \to \mathbb{Z}$ defined by $g(n) = n^2$ is *not* injective.
  - $g(-1) = 1$ and $g(1) = 1$, but $-1 \neq 1$. ✗
- $h : \mathbb{R} \to \mathbb{R}$ defined by $h(x) = \sin(x)$ is *not* injective.
  - $h(0) = h(\pi) = 0$, but $0 \neq \pi$. ✗
- $f : \mathbb{Z} \to \mathbb{Z}_5$ defined by $f(x) = x \bmod 5$ is *not* injective.
  - $f(2) = f(7) = 2$, but $2 \neq 7$. ✗

Surjective Functions

Definition 41: A function $f : A \to B$ is *surjective* (*right-total* or *onto*) if every element in the codomain is the image of at least one element in the domain. Formally:

$$\forall b \in B.\, \exists a \in A.\, f(a) = b$$

For surjective functions, $\mathrm{Range}(f) = \mathrm{Cod}(f)$, *i.e.*, there are *no "uncovered"* elements in the right side.



Examples of Surjective Functions

Key insight: A surjective function "covers" the entire codomain. Every possible output is actually achieved by some input.

Example: $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x^3$ is surjective.

- For any $y \in \mathbb{R}$, let $x = \sqrt[3]{y}$. Then $f(x) = \left(\sqrt[3]{y}\right)^3 = y$. ✓

Example: $g : \mathbb{R} \to [0, \infty)$ defined by $g(x) = x^2$ is surjective.

- For any $y \geq 0$, let $x = \sqrt{y}$. Then $g(x) = \left(\sqrt{y}\right)^2 = y$. ✓

Example: $f : \mathbb{Z} \to \mathbb{Z}_5$ defined by $f(x) = x \bmod 5$ is surjective.

- Every element $\{0, 1, 2, 3, 4\}$ is achieved: $f(0) = 0, f(1) = 1, f(2) = 2, f(3) = 3, f(4) = 4$. ✓

Example (Computer science applications):

- **Color quantization:** Map 24-bit colors to 8-bit palette (should cover all palette colors).
- **Load balancing:** Distribute requests across servers (all servers should be used).
- **Hash table design:** Hash function should potentially reach every bucket.

Examples of Non-Surjective Functions

Example (Counter-examples (not surjective)):

- $g : \mathbb{N} \to \mathbb{N}$ defined by $g(n) = 2n$ is *not* surjective.
  - Odd numbers like $3, 5, 7, \ldots$ are never achieved since $2n$ is always even. ✗
- $h : \mathbb{R} \to \mathbb{R}$ defined by $h(x) = x^2$ is *not* surjective.
  - Negative numbers like $-1, -2, -3, \ldots$ are never achieved since $x^2 \geq 0$. ✗
- $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = e^x$ is *not* surjective.
  - Negative numbers and zero are never achieved since $e^x > 0$ for all x. ✗

Bijective Functions

Definition 42: A function $f : A \to B$ is *bijective* if it is both injective and surjective. A bijective function establishes a *one-to-one correspondence* between the elements of A and B.

Equivelently, f is bijective iff it has an inverse function $f^{-1} : B \to A$.

Example: $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = 2x + 1$ is bijective.

- **Injective:** If $2x_1 + 1 = 2x_2 + 1$, then $x_1 = x_2$. ✓
- **Surjective:** For any $y \in \mathbb{R}$, let $x = \frac{y-1}{2}$. Then $f(x) = y$. ✓
- **Inverse:** $f^{-1}(y) = \frac{y-1}{2}$.

Example: $g : [0, \infty) \to [0, \infty)$ defined by $g(x) = x^2$ is bijective.

- **Injective:** If $x_1^2 = x_2^2$ and $x_1, x_2 \geq 0$, then $x_1 = x_2$. ✓
- **Surjective:** For any $y \geq 0$, let $x = \sqrt{y}$. Then $g(x) = y$. ✓
- **Inverse:** $g^{-1}(y) = \sqrt{y}$.

Examples of Bijective Functions

Key insight: Bijective functions are "perfect matchings" between sets. Every element in A pairs with *exactly one* element in B, and vice versa.

Example (Computer science applications):

- **Encryption algorithms:** Must be bijective to ensure decryption is possible.
- **Base conversion:** Bijection between decimal and binary representations.
- **Perfect hash functions:** Bijective mapping from keys to table positions.
- **Coordinate transformations:** Reversible mappings between coordinate systems.

Function Composition

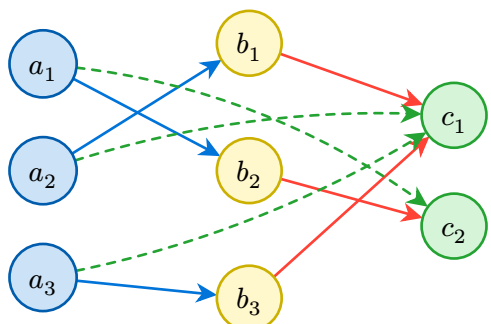
Definition 43: Let $f : A \to B$ and $g : B \to C$ be two functions. The *composition* of g and f, denoted $g \circ f$ (read as "g composed with f" or "g after f"), is a function from A to C defined by:

$$(g \circ f)(a) = g(f(a))$$

Example: Let $f : \mathbb{R} \to \mathbb{R}$ be $f(x) = x^2$ and $g : \mathbb{R} \to \mathbb{R}$ be $g(x) = x + 1$.

- $(g \circ f)(x) = g(f(x)) = g(x^2) = x^2 + 1$
- $(f \circ g)(x) = f(g(x)) = f(x + 1) = (x + 1)^2 = x^2 + 2x + 1$

Note: $g \circ f \neq f \circ g$ (composition is not commutative!)



Examples of Function Composition

Key insight: Function composition is like a "pipeline" — the output of f becomes the input to g.

Read right-to-left: $g \circ f$ means "first apply f, then apply g", or rather, "do g after f".

Example (Computer science applications):

- **Function pipelines:** `data |> filter |> map |> reduce`
- **Compiler design:** lexer $\to$ parser $\to$ optimizer $\to$ code generator
- **Data processing:** clean $\to$ transform $\to$ aggregate $\to$ visualize

Properties of Function Composition

- **Associativity:** If $f : A \to B$, $g : B \to C$, and $h : C \to D$, then $(h \circ g) \circ f = h \circ (g \circ f)$.
- The *identity* function acts as a *neutral* element for composition:
  - $\mathrm{id}_B \circ f = f$ for any function $f : A \to B$.
  - $f \circ \mathrm{id}_A = f$ for any function $f : A \to B$.
- Composition *preserves* the properties of functions:
  - If f and g are injective, so is $g \circ f$.
  - If f and g are surjective, so is $g \circ f$.
  - If f and g are bijective, so is $g \circ f$.
- Note that in general, $g \circ f \neq f \circ g$, *i.e.*, function composition is *not commutative*.

Functional Powers

Definition 44: The *functional power* of $f : A \to A$ can defined inductively:

$$f^0 = \mathrm{id}_A$$
$$f^{n+1} = f \circ f^n = f^n \circ f$$

Note: This definition also works for functions $f : X \to Y$ with $Y \subseteq X$.

Note: To avoid the confusion with exponential powers (*e.g.*, $f^n(x)$ could be interpreted as $(f(x))^n$), we can use the notation $f^{\circ n}$ to denote the n-th functional power.

Example: Let $f : \mathbb{R} \to \mathbb{R}$ be $f(x) = 2x$.

- $f^0(x) = x$ (identity)
- $f^1(x) = 2x$
- $f^2(x) = f(f(x)) = f(2x) = 2(2x) = 4x$
- $f^3(x) = f(f^2(x)) = f(4x) = 2(4x) = 8x$
- In general: $f^n(x) = 2^n x$

Examples of Functional Powers

Example: Let $g : \mathbb{Z} \to \mathbb{Z}$ be $g(x) = x + 1$ (successor function).

- $g^{\circ 0}(x) = x$
- $g^{\circ 1}(x) = x + 1$
- $g^{\circ 2}(x) = g(g(x)) = g(x + 1) = (x + 1) + 1 = x + 2$
- $g^{\circ 3}(x) = g(g^{\circ 2}(x)) = g(x + 2) = (x + 2) + 1 = x + 3$
- In general: $g^{\circ n}(x) = x + n$

Inverse Functions

Definition 45: If $f : A \to B$ is a bijective function, then its *inverse function*, denoted $f^{-1} : B \to A$, is:

$$f^{-1}(b) = a \quad \text{iff} \quad f(a) = b$$

Note: A function has an inverse *if and only if* it is bijective.

Example: Let $f : \mathbb{R} \to \mathbb{R}$ be $f(x) = 2x + 1$. We found it's bijective. To find $f^{-1}(y)$, let $y = 2x + 1$. Solving for x, we get $x = \frac{y-1}{2}$. So, $f^{-1}(y) = \frac{y-1}{2}$.

Example: The exponential function $\exp : \mathbb{R} \to (0, \infty)$ defined by $\exp(x) = e^x$ is bijective.

- Its inverse is the natural logarithm: $\ln : (0, \infty) \to \mathbb{R}$.
- We have: $\ln(e^x) = x$ for all $x \in \mathbb{R}$ and $e^{\ln y} = y$ for all $y > 0$.

Note: If we tried $\exp : \mathbb{R} \to \mathbb{R}$, it wouldn't be surjective (negative numbers never achieved), hence no inverse! The codomain restriction is essential.

Examples of Inverse Functions

Example (Polynomial with domain restriction): Consider $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x^2$.

- This is **not** injective: $f(-2) = f(2) = 4$, so no inverse exists.

However, if we **restrict the domain** to $f : [0, \infty) \to [0, \infty)$, then:

- f becomes bijective (both injective and surjective on non-negative reals).
- The inverse is $f^{-1}(y) = \sqrt{y}$ for $y \geq 0$.

Key insight: Many functions need domain/codomain restrictions to become invertible. This is crucial in calculus when finding inverse functions!

Example: The sine function $\sin : \mathbb{R} \to \mathbb{R}$ is **not** injective (it's periodic), so it has no inverse.

However, with domain restriction, $\sin : \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \to [-1, 1]$ **is** bijective, with inverse $\arcsin : [-1, 1] \to \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

Similarly: $\cos : [0, \pi] \to [-1, 1]$ has inverse arccos, and $\tan : \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \to \mathbb{R}$ has inverse arctan.

Characterization of Invertible Functions

Theorem 4: A function $f : A \to B$ has an inverse function $f^{-1} : B \to A$ *if and only if* f is bijective.

Proof: We prove both directions.

($\Rightarrow$): If f has an inverse, then f is bijective.

Assume $f^{-1} : B \to A$ is an inverse of $f : A \to B$ with the following properties:

- $f^{-1}(f(x)) = x$ for all $x \in A$
- $f(f^{-1}(y)) = y$ for all $y \in B$

We are going to show that f is both injective and surjective.

1. **f is injective:** Let $x_1, x_2 \in A$ such that $f(x_1) = f(x_2)$. Applying f^{-1} to both sides, we get:

$$f^{-1}(f(x_1)) = f^{-1}(f(x_2))$$

By the property of the inverse, this simplifies to:

$$x_1 = x_2$$

Characterization of Invertible Functions [2]

Thus, f is injective.

2. **f is surjective:** Let $y \in B$. We need to find an $x \in A$ such that $f(x) = y$. Take $x = f^{-1}(y)$. Then by the property of the inverse, we have:

$$f(x) = f(f^{-1}(y)) = y$$

Hence, f is surjective.

Therefore, f is bijective. ■

($\Leftarrow$): If f is bijective, then f has an inverse f^{-1}.

Assume $f : A \to B$ is bijective. This means:

- f is injective: $(f(x_1) = f(x_2)) \to (x_1 = x_2)$
- f is surjective: for every $y \in B$, there exists $x \in A$ such that $f(x) = y$

We need to construct an inverse function $f^{-1} : B \to A$, satisfying:

$$f^{-1}(f(x)) = x \quad \text{and} \quad f(f^{-1}(y)) = y$$

Characterization of Invertible Functions [3]

Construction of f^{-1}: For each $y \in B$:

- Since f is surjective, there exists at least one $x \in A$ such that $f(x) = y$.
- Since f is injective, this x is unique.

We define a function $f^{-1} : B \to A$ by:

$$f^{-1}(y) = \text{the unique } x \in A \text{ such that } f(x) = y$$

Verification:

1. For all $x \in A$, by definition of f^{-1}:

$$f^{-1}(f(x)) = x$$

2. For all $y \in B$, since $f^{-1}(y)$ is defined as the unique x with $f(x) = y$, we have:

$$f(f^{-1}(y)) = y$$

Therefore, f^{-1} is indeed the inverse of f. ■

Thus, a function has an inverse *if and only if* it is bijective. □

Some Properties of Inverse Functions

Theorem 5: If $f : A \to B$ is a bijective function with inverse $f^{-1} : B \to A$:

- f^{-1} is also bijective.
- $(f^{-1} \circ f)(a) = a$ for all $a \in A$ (*i.e.*, $f^{-1} \circ f = \mathrm{id}_A$).
- $(f \circ f^{-1})(b) = b$ for all $b \in B$ (*i.e.*, $f \circ f^{-1} = \mathrm{id}_B$).
- If $f : A \to B$ and $g : B \to C$ are both bijective, then $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

Monotonic Functions

Definition 46: A function $f : A \to B$ is called *monotonic* if it preserves order relationships.

For real-valued functions $f : \mathbb{R} \to \mathbb{R}$:

- *Monotonic increasing* (or *isotone*) if $x \leq y \to f(x) \leq f(y)$
- *Strictly increasing* if $x < y \to f(x) < f(y)$
- *Monotonic decreasing* (or *antitone*) if $x \leq y \to f(x) \geq f(y)$
- *Strictly decreasing* if $x < y \to f(x) > f(y)$

Note: More generally, for any posets $\langle A, \leq_A \rangle$ and $\langle B, \leq_B \rangle$:

- f is *order-preserving* (monotone) if $x \leq_A y \to f(x) \leq_B f(y)$
- f is *order-reversing* (antitone) if $x \leq_A y \to f(x) \geq_B f(y)$

Monotonicity implies Injectivity

Theorem 6: If $f : \mathbb{R} \to \mathbb{R}$ is *strictly increasing* or *strictly decreasing*, then f is *injective*.

Proof: Suppose f is strictly increasing. Let $x_1, x_2 \in \mathbb{R}$ with $x_1 \neq x_2$. Then either $x_1 < x_2$ or $x_2 < x_1$.

- If $x_1 < x_2$, then $f(x_1) < f(x_2)$ by strict monotonicity, so $f(x_1) \neq f(x_2)$.
- If $x_2 < x_1$, then $f(x_2) < f(x_1)$ by strict monotonicity, so $f(x_1) \neq f(x_2)$.

In both cases, $f(x_1) \neq f(x_2)$, proving injectivity. The proof for strictly decreasing is analogous. □

Key insights:

- Monotonic functions have *predictable* behavior: they never "change direction"
- *Strictly monotonic* functions are *always injective* (one-to-one)
- Non-strict monotonic functions may have "flat" regions (also known as "*plateaus*") where different inputs map to the same output
- Fun fact: monotonic functions are *order-homomorphisms* between posets!

Examples of Monotonic Functions

Examples: *Strictly monotonic* functions (which are always injective):

- $f(x) = 2x + 1$ is strictly increasing on $\mathbb{R}$, hence injective. ✓
- $g(x) = -x$ is strictly decreasing on $\mathbb{R}$, hence injective. ✓
- $h(x) = x^3$ is strictly increasing on $\mathbb{R}$, hence injective. ✓
- $k(x) = e^x$ is strictly increasing on $\mathbb{R}$, hence injective. ✓

Examples: *Monotonic* but *NOT strictly* monotonic functions (not necessarily injective):

- $f(x) = \lfloor x \rfloor$ is monotonic increasing but NOT strictly increasing.
  - For $x \in [2, 3)$, we have $f(x) = 2$ (constant on intervals).
  - This is NOT injective: $f(2.1) = f(2.9) = 2$ but $2.1 \neq 2.9$. ✗
- $g(x) = \begin{cases} x & \text{if } x \leq 0 \\ 0 & \text{if } 0 < x < 1 \\ x-1 & \text{if } x \geq 1 \end{cases}$ is monotonic but not injective.

Example: $f : (\mathbb{R} \setminus \{0\}) \to \mathbb{R}$ defined by $f(x) = \frac{1}{x}$ is **injective but not monotonic**.

- If $f(x_1) = f(x_2)$, then $\frac{1}{x_1} = \frac{1}{x_2}$ implies $x_1 = x_2$. ✓
- However, f is not monotonic since it decreases on $(-\infty, 0)$ and $(0, \infty)$, but $f(-1) < f(1)$. ✗

Function Properties Overview

Functions can be characterized by several key properties that determine their mathematical behavior.

Property	Definition
Functional (Right-unique)	Each input maps to *at most one* output
Total (Left-total)	Each input maps to *at least one* output (defined everywhere)
Partial	Functional but not total (may be undefined for some inputs)
Injective (Left-unique)	Different inputs $\Longrightarrow$ different outputs
Surjective (Right-total)	Every codomain element is covered
Bijective	Both injective and surjective (one-to-one correspondence)
Monotonic	Preserves or reverses order relationships
Continuous	Small input changes $\Longrightarrow$ small output changes

Characteristic Functions

Definition 47: For any set $S \subseteq A$, the *characteristic function* (or *indicator function*) $\chi_S : A \to \{0, 1\}$ is:

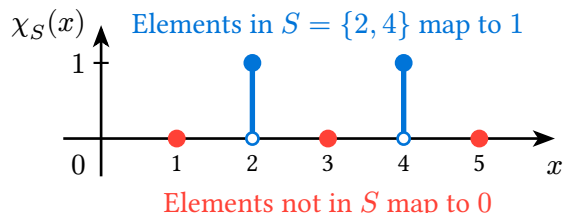
$$\chi_S(x) = \begin{cases} 1 \text{ if } x \in S \\ 0 \text{ if } x \notin S \end{cases}$$

This function "indicates" membership in set S.

Example: Let $A = \{1, 2, 3, 4, 5\}$ and $S = \{2, 4\}$. Then χ_S maps:

- $\chi_S(1) = 0$ (since $1 \notin S$)
- $\chi_S(2) = 1$ (since $2 \in S$)
- $\chi_S(3) = 0$ (since $3 \notin S$)
- $\chi_S(4) = 1$ (since $4 \in S$)
- $\chi_S(5) = 0$ (since $5 \notin S$)

So $\chi_S = \{(1, 0), (2, 1), (3, 0), (4, 1), (5, 0)\}$.



Properties of Characteristic Functions

Definition 48: For sets $A, B \subseteq U$:

- $\chi_{A \cap B} = \chi_A \cdot \chi_B$ (pointwise multiplication)
- $\chi_{A \cup B} = \chi_A + \chi_B - \chi_A \cdot \chi_B$
- $\chi_{\overline{A}} = 1 - \chi_A$ (complement)
- $\chi_{A \triangle B} = \chi_A + \chi_B - 2\chi_A \cdot \chi_B$ (symmetric difference)
- $\chi_\emptyset = 0$ and $\chi_U = 1$ (constant functions)

Example (Applications):

- **Probability theory:** Indicator random variables
- **Database queries:** Boolean conditions in WHERE clauses
- **Set operations:** Converting logical operations to arithmetic
- **Machine learning:** Feature encoding (one-hot encoding)
- **Computer graphics:** Masking and selection operations
- **Digital signal processing:** Window functions and filters

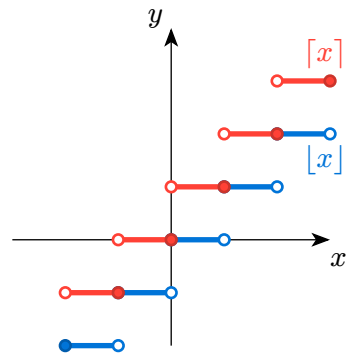
Floor and Ceiling Functions

Definition 49:

- *Floor function* $\lfloor$: $\mathbb{R} \to \mathbb{Z}$ maps x to the largest integer $\leq x$: $\lfloor x \rfloor = \max\{n \in \mathbb{Z} \mid n \leq x\}$
- *Ceiling function* $\lceil$: $\mathbb{R} \to \mathbb{Z}$ maps x to the smallest integer $\geq x$: $\lceil x \rceil = \min\{n \in \mathbb{Z} \mid n \geq x\}$

Example:

x	$\lfloor x \rfloor$	$\lceil x \rceil$	**Note**
3.7	3	4	Positive non-integer
-2.3	-3	-2	Negative non-integer
5	5	5	Integer (floor = ceiling)
0	0	0	Zero
-1	-1	-1	Negative integer



Summary: Functions

Functions establish relationships between sets where every input has exactly one output:

- *Injective:* Different inputs $\rightarrow$ Different outputs (one-to-one)
- *Surjective:* Every output is achieved (onto)
- *Bijective:* Both injective and surjective (perfect correspondence)

Key applications in computer science:

- **Database design:** Functions model relationships
- **Algorithm analysis:** Function properties affect complexity
- **Cryptography:** Bijective functions enable encryption/decryption
- **Type systems:** Function signatures and type hierarchies
- **Data structures:** Mappings, hashing, and uniqueness constraints

Cardinality & Infinity

"God made the integers, all else is the work of man."

— Leopold Kronecker

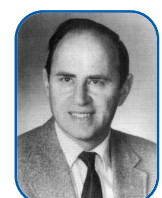
Giuseppe Peano

Leopold Kronecker

David Hilbert

Kurt Gödel

John von Neumann



Paul Cohen

Size of Sets

Definition 50: The *size* of a *finite* set X, denoted $|X|$, is the number of elements it contains.

Examples:

- Let $A = \{$🪐, 🦕, 🎻$\}$, then $|A| = 3$, since A contains *exactly 3* elements.
- Let $B = \{$🥝, 🥝, 🥝$\}$, then $|B| = 1$, since B contains *only one unique* element (the kiwi).
- $|\mathcal{P}(\{1, 2,$ 🐈$\})| = 2^3 = 8$, since the power set consists of *all 8 possible subsets* of $\{1, 2,$ 🐈$\}$.
- $|\emptyset| = 0$, since the *empty* set contains *no elements*.
- $|\mathbb{N}| = \infty$, since there are *infinitely many* natural numbers.
- $|\mathbb{R}| = \infty$, since there are *infinitely many* real numbers.

Cardinality of Sets

Definition 51: The *cardinality* of a set X, denoted $|X|$, is a measure of its "size".

- For *finite* sets, cardinality $|X|$ is the same as *size*, *i.e.*, the number of elements in X.
- For *infinite* sets, cardinality $|X|$ describes the "type" of infinity, distinguishing between *countable* (like $\mathbb{N}$) and *uncountable* (like $\mathbb{R}$) infinities.

Example (Finite sets):

- $|\{a, b, c\}| = 3$ (three distinct elements)
- $|\emptyset| = 0$ (empty set has no elements)
- $|\mathcal{P}(\{1, 2\})| = |\{\emptyset, \{1\}, \{2\}, \{1, 2\}\}| = 4 = 2^2$

Example (Infinite sets):

- $|\mathbb{N}| = \aleph_0$ ("aleph-null" — smallest infinite cardinal, denoting *countable* infinity)
- $|\mathbb{Z}| = \aleph_0$ (same size as natural numbers!)
- $|\mathbb{Q}| = \aleph_0$ (rationals are also *countably* infinite)
- $|\mathbb{R}| = 2^{\aleph_0} = \mathfrak{c}$ ("continuum" — *uncountably* infinite)
- $|\mathcal{P}(\mathbb{N})| = 2^{\aleph_0} > \aleph_0$ (power set is always "larger")

Cardinal Numbers

Note: $|X|$ is *not* just a number, but a *cardinal number*.

- Cardinal numbers extend natural numbers to describe sizes of infinite sets.
- The *finite* cardinal numbers are just natural numbers: $0, 1, 2, 3, \ldots$.
- The first (smallest) *infinite* cardinal is $\aleph_0$ (the cardinality of $\mathbb{N}$).
- *Arithmetic* operations on cardinal numbers *differ* from those on natural numbers.
  - For example, $\aleph_0 + 1 = \aleph_0$ and $\aleph_0 \cdot 2 = \aleph_0$.

Equinumerosity

Definition 52: Two sets A and B have the same *cardinality* and are called *equinumerous*, denoted $|A| = |B|$ or $A \approx B$, iff there exists a *bijection* (one-to-one correspondence) between A and B.

Intuition: If you can pair up every element of A with exactly one element of B, with nothing left over on either side, then A and B have the same cardinality.

Theorem 7: Equinumerosity is an equivalence relation.

Proof: Let A, B, C be sets.

- *Reflexivity:* The identity map $\mathrm{id}_A : A \to A$, where $\mathrm{id}_A(x) = x$, is a bijection, so $A \approx A$.
- *Symmetry:* Suppose $A \approx B$, then there is a bijection $f : A \to B$. Since it is a bijection, its inverse f^{-1} exists and is also a bijection. Hence, $f^{-1} : B \to A$ is a bijection, so $B \approx A$.
- *Transitivity:* Suppose that $A \approx B$ and $B \approx C$, *i.e.*, there are bijections $f : A \to B$ and $g : B \to C$. Then the composition $g \circ f : A \to C$ is also a bijection. So $A \approx C$.

Therefore, equinumerosity is an equivalence relation. $\square$

Countable Sets

Definition 53: A set is *countable* if it is either:

1. *Finite*, or
2. Has the same cardinality as $\mathbb{N}$ (*i.e.*, there exists a *bijection* with $\mathbb{N}$)

Intuition: A countable set is one whose elements can be "listed" or "enumerated" in a sequence.

Note: When an infinite set is countable, its cardinality is $\aleph_0$ (*"aleph-null"*).

Coming up: We'll see *different* ways to characterize countable sets (using bijections, injections, or enumerations). They're all *equivalent* — use whichever is most convenient for your proof!

Example (Basic countable sets):

- $\mathbb{N}_{\text{odd}} = \{1, 3, 5, 7, \ldots\}$ with bijection $f : \mathbb{N} \to \mathbb{N}_{\text{odd}}$ defined by $f(n) = 2n + 1$
- $\mathbb{N}_{\text{even}} = \{0, 2, 4, 6, \ldots\}$ with bijection $f : \mathbb{N} \to \mathbb{N}_{\text{even}}$ defined by $f(n) = 2n$
- Prime numbers $\{2, 3, 5, 7, 11, 13, \ldots\}$ are countable (can be enumerated)

Examples of Countable Sets

Key insight: Two sets have the same cardinality if we can pair up their elements perfectly — establishing a one-to-one correspondence (bijection) between them.

Example: The set of *integers* $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$ is *countable*: $|\mathbb{Z}| = \aleph_0$

Bijection $f : \mathbb{N} \to \mathbb{Z}$ is defined by:

$$f(n) = \begin{cases} \frac{n}{2} & \text{if } n \text{ is even} \\ -\frac{n+1}{2} & \text{if } n \text{ is odd} \end{cases} \qquad \begin{bmatrix} \mathbb{N}: & 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 & \ldots \\ \mathbb{Z}: & 0 & -1 & 1 & -2 & 2 & -3 & 3 & -4 & \ldots \end{bmatrix}$$

This systematically pairs each natural number with exactly one integer, covering all integers exactly once.

Properties of Countable Sets

Theorem 8 (Properties of countable sets):

1. Any subset of a countable set is countable
2. The union of countably many countable sets is countable
3. The Cartesian product of two countable sets is countable ($\mathbb{N} \times \mathbb{N}$ is countable)
4. The set of finite sequences over a countable alphabet is countable

Example (More countable sets):

- **Rational numbers** $\mathbb{Q}$: Can be enumerated by listing fractions $\frac{p}{q}$ systematically
- **Finite binary strings** $\{0, 1, 00, 01, 10, 11, 000, \ldots\}$: Countably infinite
- **Polynomials with integer coefficients**: Can be systematically enumerated
- **Finite subsets of** $\mathbb{N}$: Each finite subset can be encoded as a finite binary string

Dedekind-Infinite Sets

Definition 54: A set X is *Dedekind-infinite* if some proper subset $Y \subset X$ is equinumerous to it, *i.e.*, there is a bijection between X and one of its proper subsets.

A set that is not Dedekind-infinite is called *Dedekind-finite*.

Note: Intuitively, an "infinite" set can be put in one-to-one correspondence with a part of itself. This is impossible for finite sets.

Example: The set of natural numbers $\mathbb{N}$ is Dedekind-infinite:

- Let $Y = \{2, 4, 6, 8, \ldots\} = \mathbb{N}_{\text{even}} \subset \mathbb{N}$ (proper subset)
- Define $f : \mathbb{N} \to Y$ by $f(n) = 2n$ (bijection)
- Since $\mathbb{N} \approx \mathbb{N}_{\text{even}}$, the set $\mathbb{N}$ is Dedekind-infinite

Examples of Dedekind-Infinite Sets

Examples:

- $\mathbb{Z}$ is Dedekind-infinite: bijection with positive integers
- $\mathbb{Q}$ is Dedekind-infinite: bijection with positive rationals
- $\mathbb{R}$ is Dedekind-infinite: bijection with $(0, 1)$

Key insight: Being Dedekind-infinite is equivalent to being infinite (assuming the Axiom of Choice). This gives us a purely set-theoretic definition of infinity without reference to natural numbers.

Alternative Characterization: Injection into $\mathbb{N}$

If we can "fit" all elements of S into the natural numbers without overlaps, then S is countable.

Definition 55: A set S is *countable* if there exists an *injection* $f : S \to \mathbb{N}$.

This means each element of S can be assigned a unique natural number.

Note: This characterization works both for *finite* and countably infinite sets, and is often easier to use than constructing a full bijection!

Infinite Subsets of Natural Numbers

Theorem 9: Any infinite subset $A \subseteq \mathbb{N}$ has a bijection with $\mathbb{N}$.

Proof: Let $A \subseteq \mathbb{N}$ be infinite. We construct a bijection $f : \mathbb{N} \to A$ recursively as follows:

- $f(0) = a_0 = \min A$
- $f(n+1) = \min(A \setminus \{f(0), f(1), ..., f(n)\})$

By well-ordering principle, each subset of $\mathbb{N}$, has a least element. At each step, $A \setminus \{f(0), ..., f(n)\}$ is still infinite (otherwise A would be finite), so it has a minimum element, making the construction well-defined.

Properties of f:

- **Injective:** By construction, $f(i) \neq f(j)$ for $i \neq j$, since we explicitly exclude previous values.
- **Surjective:** Every $a \in A$ eventually becomes the minimum of a remaining set, so $a = f(n)$ for some n.

Therefore, f is a bijection between $\mathbb{N}$ and A. Thus $A \approx \mathbb{N}$, meaning $|A| = \aleph_0$. $\square$

Key insight: There is essentially only "one size" of countably infinite sets, namely $\aleph_0$.

Enumerable Sets

Definition 56: A set X is *enumerable* if there is a surjection $e : \mathbb{N} \to X$.

The function e is called an *enumeration* of X, and we say that e *enumerates* X.

Intuition: An enumerable set is one whose elements can be "listed" (possibly with repetitions) by a function from natural numbers: $e(0), e(1), e(2), e(3), \ldots$

Note: Each element of X appears at least once in this list, though some elements might appear multiple times, since the enumeration requires only a *surjection*, not a bijection.

Three Faces of Countability

The following characterizations of countable sets are all **equivalent**:

Theorem 10: For any set X, the following are equivalent:

1. X is *enumerable*: there exists a surjection $e : \mathbb{N} \to X$
2. X is *empty*, or there exists an injection $f : X \to \mathbb{N}$
3. X is *countable*: X is finite or has a bijection with $\mathbb{N}$

What this means: All three perspectives capture the *same* concept! You can use whichever characterization is most convenient for your specific proof.

- **Bijection** ($1 \leftrightarrow 1$): Perfect pairing, shows exact size
- **Injection** ($1 \to 1$): Fitting into $\mathbb{N}$ without collisions
- **Surjection** (enumeration): Listing all elements (possibly with repetitions)

Three Faces of Countability [2]

Proof: We prove $(1) \Rightarrow (2) \Rightarrow (3) \Rightarrow (1)$.

$(1) \Rightarrow (2)$: Suppose $e : \mathbb{N} \to X$ is a surjection.

If $X = \emptyset$, we're done. Otherwise, X is non-empty.

For each $x \in X$, define $f(x) = \min\{n \in \mathbb{N} \mid e(n) = x\}$ (the first index where x appears in the enumeration). This is well-defined because e is surjective, so every x has at least one preimage.

f is injective: if $f(x) = f(y) = n$, then $e(n) = x$ and $e(n) = y$, so $x = y$.

$(2) \Rightarrow (3)$: Suppose $X = \emptyset$ or there exists an injection $f : X \to \mathbb{N}$.

- If $X = \emptyset$, then X is finite (hence countable).
- Otherwise, f maps X injectively into $\mathbb{N}$, so $X \approx \mathrm{Im}(f) \subseteq \mathbb{N}$, where $\mathrm{Im}(f) = f(X)$.
- By Theorem 9 about infinite subsets of $\mathbb{N}$, either $\mathrm{Im}(f)$ is finite or $\mathrm{Im}(f) \approx \mathbb{N}$.
- In either case, X is countable.

$(3) \Rightarrow (1)$: Suppose X is countable.

- If X is finite with n elements, list them as $x_0, x_1, \ldots, x_{n-1}$, and define:

Three Faces of Countability [3]

$$e(k) = \begin{cases} x_k & \text{if } k < n \\ x_0 & \text{if } k \geq n \end{cases}$$

This is a surjection from $\mathbb{N}$ to X (repeating x_0 for all $k \geq n$).

- If $X \approx \mathbb{N}$, let $g : \mathbb{N} \to X$ be a bijection. Then g is also a surjection.

In both cases, we have constructed a surjection $e : \mathbb{N} \to X$. $\square$

Practical guide — when to use which characterization:

- **Bijection** $f : \mathbb{N} \to X$: When you want to (or can) show sets have *exactly the same size* or construct an explicit listing
- **Injection** $f : X \to \mathbb{N}$: When X is a *subset* or embeds naturally into $\mathbb{N}$ (often easiest!)
- **Surjection** $e : \mathbb{N} \to X$: When you can *enumerate* elements algorithmically (allows repetitions)

Key insight: You can prove countability using *any* of these three methods, and then freely use properties from *all* three perspectives in subsequent arguments!

Examples of Enumerable Sets

Example (Explicit enumerations):

Even numbers: $e(n) = 2n$ gives enumeration $0, 2, 4, 6, 8, \ldots$

Odd numbers: $e(n) = 2n + 1$ gives enumeration $1, 3, 5, 7, 9, \ldots$

Perfect squares: $e(n) = n^2$ gives enumeration $0, 1, 4, 9, 16, 25, \ldots$

Prime numbers: $e(0) = 2, e(1) = 3, e(2) = 5, e(3) = 7, e(4) = 11, \ldots$

- **Note:** while we don't have a closed formula, we can enumerate primes algorithmically (*e.g.*, using the Sieve of Eratosthenes).

Examples of Enumerable Sets [2]

Example (Integers enumeration): The enumeration $e : \mathbb{N} \to \mathbb{Z}$ defined by:

$$e(n) = \begin{cases} \frac{n}{2} & \text{if } n \text{ is even} \\ -\frac{n+1}{2} & \text{if } n \text{ is odd} \end{cases}$$

produces the sequence: $0, -1, 1, -2, 2, -3, 3, -4, 4, -5, 5, \ldots$

This systematically alternates between non-negative and negative integers.

Example (Finite strings): The set Σ^* of all finite strings over alphabet $\Sigma = \{a, b\}$ can be enumerated by:

- First, list strings by length
- Within each length, use lexicographic order

Enumeration: $\varepsilon, a, b, aa, ab, ba, bb, aaa, aab, aba, abb, baa, \ldots$

This shows Σ^* is countable for any finite alphabet Σ.

Examples of Enumerable Sets [3]

Example (Rational numbers): We can enumerate *positive* rationals by listing fractions $\frac{p}{q}$ ordered by $p + q$:

$p + q$	Fractions	Reduced	Indices	Count
1	$\frac{0}{1}$	—	—	0
2	$\frac{1}{1}$	$\frac{1}{1}$	0	1
3	$\frac{1}{2}, \frac{2}{1}$	$\frac{1}{2}, \frac{2}{1}$	1, 2	2
4	$\frac{1}{3}, \frac{2}{2}, \frac{3}{1}$	$\frac{1}{3}, \frac{3}{1}$	3, 4	2
5	$\frac{1}{4}, \frac{2}{3}, \frac{3}{2}, \frac{4}{1}$	$\frac{1}{4}, \frac{2}{3}, \frac{3}{2}, \frac{4}{1}$	5, 6, 7, 8	4
…	…	…	…	…

Skip non-reduced fractions (like $\frac{2}{2}$) and enumerate: $\frac{1}{1}, \frac{1}{2}, \frac{2}{1}, \frac{1}{3}, \frac{3}{1}, \frac{1}{4}, \frac{2}{3}, \frac{3}{2}, \frac{4}{1}, \ldots$

Adding zero and negative rationals by interleaving gives full enumeration of $\mathbb{Q}$.

Zig-Zag Enumeration

Theorem 11: $\mathbb{N}^2$ is countable.

Proof: We enumerate all pairs in $\mathbb{N}^2$ by traversing diagonals of constant sum $n + k$.

$n \downarrow$ $k \rightarrow$	**0**	**1**	**2**	**3**	**...**
0	$\langle 0, 0 \rangle$	$\langle 0, 1 \rangle$	$\langle 0, 2 \rangle$	$\langle 0, 3 \rangle$	...
	0	2	5	9	
1	$\langle 1, 0 \rangle$	$\langle 1, 1 \rangle$	$\langle 1, 2 \rangle$	$\langle 1, 3 \rangle$	...
	1	4	8	13	
2	$\langle 2, 0 \rangle$	$\langle 2, 1 \rangle$	$\langle 2, 2 \rangle$	$\langle 2, 3 \rangle$	...
	3	7	12	18	
3	$\langle 3, 0 \rangle$	$\langle 3, 1 \rangle$	$\langle 3, 2 \rangle$	$\langle 3, 3 \rangle$	...
	6	11	17	24	
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$

Zig-Zag Enumeration [2]

Enumeration order: We traverse (upward) diagonals where $n + k = s$ for $s = 0, 1, 2, \ldots$:

- Diagonal $s = 0$: $\langle 0, 0 \rangle$ (index 0)
- Diagonal $s = 1$: $\langle 1, 0 \rangle, \langle 0, 1 \rangle$ (indices 1–2)
- Diagonal $s = 2$: $\langle 2, 0 \rangle, \langle 1, 1 \rangle, \langle 0, 2 \rangle$ (indices 3–5)
- Diagonal $s = 3$: $\langle 3, 0 \rangle, \langle 2, 1 \rangle, \langle 1, 2 \rangle, \langle 0, 3 \rangle$ (indices 6–9)
- And so on...

Bijection formula: The pair $\langle n, k \rangle$ maps to index:

$$f(n, k) = \underbrace{\frac{(n+k)(n+k+1)}{2}}_{\text{pairs before diagonal } n+k} + \underbrace{k}_{\text{position within diagonal}}$$

This is the *Cantor pairing function*, which gives a bijection $f : \mathbb{N}^2 \to \mathbb{N}$.

Since we have an explicit bijection, $\mathbb{N}^2 \approx \mathbb{N}$, so $\left|\mathbb{N}^2\right| = \aleph_0$. $\square$

Zig-Zag Enumeration [3]

Key insight: The *zig-zag method* systematically visits every pair exactly once. Each diagonal is finite, and we process diagonals in order, ensuring every pair eventually appears.

Why this matters: This construction shows that "2-dimensional infinity" ($\mathbb{N}^2$) is the same size as "1-dimensional infinity" ($\mathbb{N}$). More generally, $\mathbb{N}^n$ is countable for any finite n.

Theorem 12: $\mathbb{Q}$ is countable.

Proof: Enumerate positive reduced fractions p/q ordered by $p + q$ and increasing p; skip non-reduced. Interleave 0 and negatives. This yields *enumeration*, hence $\mathbb{Q} \approx \mathbb{N}$. $\square$

Pairing Functions

Definition 57: A function $f : A \times B \to \mathbb{N}$ is an arithmetical *pairing function* if it is injective.

We say that f *encodes* $A \times B$, and that $f(a, b)$ is the *code* of the pair $\langle a, b \rangle$.

Example: The *Cantor pairing function* $g : \mathbb{N}^2 \to \mathbb{N}$ is defined as:

$$g(n, k) = \frac{(n + k + 1)(n + k)}{2} + n$$

Uncountable Sets

Definition 58: A set is *uncountable* if it is infinite but not countable.

In other words, there is *no* bijection between the set and $\mathbb{N}$.

Strategy for proving uncountability: *Cantor's diagonal argument*.

Given a list of elements, say $x_1, x_2, \ldots$ (enumerated by natural numbers), we construct a *new* element that *differs* from each x_i, showing that the list is incomplete.

Theorem 13: The set $\mathbb{B}^\omega$ of all infinite binary sequences is *uncountable*.

Proof: $\mathbb{B}^\omega$ contains sequences like 000000..., 010101..., 111000..., *etc.*

Suppose for contradiction that $\mathbb{B}^\omega$ is countable. Then we can *enumerate* its elements as $x_1, x_2, x_3, \ldots$, where each x_i is an infinite sequence of bits.

Uncountable Sets [2]

Sequence	Position 1	Position 2	Position 3	Position 4	...
x_1	$\boldsymbol{b_{11}}$	b_{12}	b_{13}	b_{14}	$\dots$
x_2	b_{21}	$\boldsymbol{b_{22}}$	b_{23}	b_{24}	$\dots$
x_3	b_{31}	b_{32}	$\boldsymbol{b_{33}}$	b_{34}	$\dots$
x_4	b_{41}	b_{42}	b_{43}	$\boldsymbol{b_{44}}$	$\dots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$
$\boldsymbol{\Delta}$	$\boldsymbol{\overline{b}_{11}}$	$\boldsymbol{\overline{b}_{22}}$	$\boldsymbol{\overline{b}_{33}}$	$\boldsymbol{\overline{b}_{44}}$	$\dots$

We construct a new sequence $\Delta = \left(\overline{b}_{11}, \overline{b}_{22}, \overline{b}_{33}, \overline{b}_{44}, \dots\right)$ where:

$$\overline{b}_{ii} = \begin{cases} 1 \text{ if } b_{ii} = 0 \\ 0 \text{ if } b_{ii} = 1 \end{cases}$$

Key observation: The diagonal elements $(b_{11}, b_{22}, b_{33}, b_{44}, \dots)$ are highlighted in red. The new sequence Δ (shown in blue) flips each diagonal element.

Uncountable Sets [3]

Why Δ is not in the list:

- $\Delta \neq x_1$ because they differ at position 1: $\overline{b}_{11} \neq b_{11}$
- $\Delta \neq x_2$ because they differ at position 2: $\overline{b}_{22} \neq b_{22}$
- ...
- In general, $\Delta \neq x_i$ for any i because they differ at position i

Since $\Delta \in \mathbb{B}^\omega$ but Δ is not in our supposedly complete enumeration, we have reached a contradiction.

Therefore, $\mathbb{B}^\omega$ is *uncountable*. $\square$

Notice the similarity to the zig-zag enumeration! Both use a 2D table structure:

- **Zig-zag:** Systematically visits all pairs (n, k) to show $\mathbb{N}^2$ is countable.
- **Diagonal argument:** Shows we *cannot* systematically visit all infinite sequences.

The key difference: zig-zag works because each diagonal is *finite*, while the diagonal argument shows that infinite sequences cannot be fully enumerated.

Examples of Uncountable Sets

Example (Important uncountable sets):

- **Real numbers** $\mathbb{R}$: Can be put in bijection with $\mathbb{B}^\omega$ via binary expansions
- **Power set** $\mathcal{P}(\mathbb{N})$: By Cantor's theorem, always larger than the original set
- **Irrational numbers** $\mathbb{I} = \mathbb{R} \setminus \mathbb{Q}$: Since $\mathbb{Q}$ is countable but $\mathbb{R}$ is not
- **Transcendental numbers**: Numbers like π, e that are not algebraic
- **Function spaces** $\mathbb{R}^\mathbb{R}$: All functions from reals to reals
- **Infinite binary strings** $\mathbb{B}^\omega$: As proved above

Example (Computer science connections):

- **Undecidable problems**: There are uncountably many languages, but only countably many algorithms
- **Real computation**: Continuous functions vs discrete algorithms
- **Cryptographic keys**: Key spaces can be countable or uncountable depending on the system
- **Information theory**: Continuous vs discrete information sources

Sets of Different Sizes

Definition 59: The cardinality of a set A is *less than or the same* as the cardinality of a set B, denoted $|A| \leq |B|$ or $A \preceq B$, if there is an *injection* (one-to-one function) from A to B.

Definition 60: Set A is *smaller* than B, denoted $|A| < |B|$ or $A \prec B$, iff there is an *injection*, but *no bijection* from A to B, *i.e.*, $A \preceq B$ and $A \not\approx B$.

Note: Using this notation, we can say that a set X is *countable* iff $X \preceq \mathbb{N}$, and *uncountable* iff $\mathbb{N} \prec X$.

Example: $\{1, 2\} \prec \{a, b, c\}$, since there is an injection $f : \{1, 2\} \to \{a, b, c\}$ defined by $f(1) = a$ and $f(2) = b$, but no bijection exists.

Example: $\mathbb{N} \preceq \mathbb{Z}$, since there is bijection (and thus an injection) $f : \mathbb{N} \to \mathbb{Z}$.

Example: $\mathbb{Z} \preceq \mathbb{N}$, since there is bijection (and thus an injection) $f : \mathbb{Z} \to \mathbb{N}$.

Example: $\mathbb{N} \prec \mathcal{P}(\mathbb{N})$, since there is an injection $f(x) = \{x\}$, but no bijection exists.

Cantor's Theorem

Theorem 14 (Cantor): $A \prec \mathcal{P}(A)$ for any set A.

Proof: The map $f(x) = \{x\}$ is an injection from A to $\mathcal{P}(A)$, since if $x \neq y$, then $\{x\} \neq \{y\}$, and so $f(x) \neq f(y)$. So we have shown that $A \preceq \mathcal{P}(A)$.

To show that $A \not\approx \mathcal{P}(A)$, suppose for reductio that there is a bijection $g : A \to \mathcal{P}(A)$.

- Consider $D = \{x \in A \mid x \notin g(x)\}$. Note that $D \subseteq A$, so $D \in \mathcal{P}(A)$.
- Since g is surjective, there exists $y \in A$ such that $g(y) = D$.
  - If $y \in D$, then by definition of D, we have $y \notin g(y)$, *i.e.*, $y \notin D$. Contradiction!
  - If $y \notin D$, then $y \notin g(y)$, so by definition of D, we have $y \in D$. Contradiction!
- Therefore, no bijection $A \to \mathcal{P}(A)$ can exist, so $A \not\approx \mathcal{P}(A)$. $\square$

Profound implication: There is no "largest" infinity! We can always construct a bigger one using the power set operation: $\aleph_0 < 2^{\aleph_0} < 2^{2^{\aleph_0}} < \ldots$

Schröder–Bernstein Theorem

Theorem 15: If $A \preceq B$ and $B \preceq A$, then $A \approx B$.

Equivalently, if there are injections $f : A \to B$ and $g : B \to A$, then there exists a bijection $h : A \to B$.

Proof: Let $f : A \to B$ and $g : B \to A$ be injections.

We write g^{-1} to denote the inverse map defined on the image $g(B)$, *i.e.*, $g^{-1} : g(B) \to B$, where $g^{-1}(a)$ denotes the unique $b \in B$ such that $g(b) = a$.

Step 1: Define inductively the sets $A_n \subseteq A$ and $B_n \subseteq B$ as follows:

- $A_0 := A \setminus g(B)$ (elements of A not in the image of g)
- $B_n := f(A_n)$ for $n \geq 0$ (images of A_n under f)
- $A_{n+1} := g(B_n)$ for $n \geq 0$ (images of B_n under g)

Define $A_\infty \subseteq A$ as the union of all A_n:

$$A_\infty = \bigcup_{n \geq 0} A_n$$

Schröder–Bernstein Theorem [2]

Step 2: Define the candidate bijection $h : A \to B$ by:

$$h(a) = \begin{cases} f(a) & \text{if } a \in A_\infty \\ g^{-1}(a) & \text{if } a \in A \setminus A_\infty \end{cases}$$

Step 3: Check that h is well-defined.

- If $a \in A_\infty$, then $h(a) = f(a)$ is well-defined since f is defined on all of A.
- If $a \in A \setminus A_\infty$, then $a \notin A_0$ (since $A_0 \subseteq A_\infty$), so $a \in g(B)$. Thus, $h(a) = g^{-1}(a)$ exists and is unique

We also need to check the two cases map into disjoint subsets of B. Observe:

$$f(A_\infty) = \bigcup_{n \geq 0} f(A_n) = \bigcup_{n \geq 0} B_n$$

If $a \notin A_\infty$ and $b = g^{-1}(a)$, then $b \notin \bigcup_{n \geq 0} B_n$, since if $b \in B_n$ for some n, then $a = g(b) \in g(B_n) = A_{n+1} \subseteq A_\infty$, contradicting $a \notin A_\infty$. Hence $g^{-1}(A \setminus A_\infty) \subseteq B \setminus f(A_\infty)$. So two image-sets $f(A_\infty)$ and $g^{-1}(A \setminus A_\infty)$ are disjoint. That prevents ambiguity across the two cases.

Schröder–Bernstein Theorem [3]

Step 4: Show that h is injective.

Take $a, a' \in A$ with $h(a) = h(a')$. Consider cases:

- **Both** $a, a' \in A_\infty$, then $f(a) = f(a')$. Since f is injective, $a = a'$.
- **Both** $a, a' \notin A_\infty$, then $g^{-1}(a) = g^{-1}(a')$. Since g is injective, $a = a'$.
- **One** in A_∞ and the **other is not**. Suppose $a \in A_\infty$ and $a' \notin A_\infty$. Then $h(a) = f(a) \in f(A_\infty)$, but $h(a') = g^{-1}(a') \in B \setminus f(A_\infty)$. These cannot be equal, so this case is impossible.

Step 5: Show that h is surjective.

Let $b \in B$. Consider two cases:

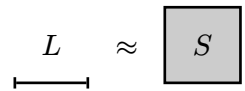
- If $b \in f(A_\infty)$, then $b = f(a)$ for some $a \in A_\infty$, and then $h(a) = f(a) = b$.
- If $b \notin f(A_\infty)$, then $a = g(b) \in A$. We claim that $a \notin A_\infty$. Otherwise, $b \in B_n$ for some n, contradicting $b \notin f(A_\infty)$. Thus $a \in A \setminus A_\infty$, so $h(a) = g^{-1}(a) = b$.

Conclusion: Since h is both injective and surjective, h is a bijection, so $A \approx B$. $\square$

Another Cantor's Theorem

Let L be the unit line, *i.e.*, the set of points $[0; 1]$. Let S be the unit square, *i.e.*, the set of points $L \times L$.

Theorem 16: $L \approx S$.



Proof[4]: Consider the function $f : L \to S$ defined by $f(x) = \langle x, x \rangle$. This is an injection, since if $f(a) = f(b)$, then $\langle a, a \rangle = \langle b, b \rangle$, so $a = b$. Thus, $L \preceq S$.

Now consider the function $g : S \to L$ that maps $\langle x, y \rangle$ to the real number obtained by *interleaving* the decimal expansions of x and y.

$$\left.\begin{aligned} x &= 0.x_1x_2x_3\ldots \\ y &= 0.y_1y_2y_3\ldots \end{aligned}\right\} \quad g(x, y) = 0.x_1y_1x_2y_2x_3y_3\ldots$$

This is an injection, since if $g(a, b) = g(c, d)$, then $a_n = c_n$ and $b_n = d_n$ for all $n \in \mathbb{N}$, so $\langle a, b \rangle = \langle c, d \rangle$. Thus, $S \preceq L$.

By Schröder–Bernstein (Theorem 15), we have that $L \approx S$. $\square$

[4] See https://math.stackexchange.com/a/183383 for more detailed analysis.

Summary: Cardinality & Infinity

Cardinality measures set "size," revealing surprising facts about infinity:

- *Finite sets:* Count elements normally
- *Countable* infinity ($\aleph_0$): Sets like $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$ that can be "listed" (enumerated)
- *Uncountable* infinity: Sets like $\mathbb{R}$, $\mathcal{P}(\mathbb{N})$ that are "too big" to list (enumerate with $\mathbb{N}$)
- **Cantor's insight:** $|A| < |\mathcal{P}(A)|$ always — there's no "largest" infinity!

Key techniques:

- *Bijections* show equal cardinality
- *Injections* show one set is no larger than another
- *Cantor's diagonal argument* proves uncountability
- *Schröder–Bernstein theorem* combines injections to yield bijections

Large Cardinal Numbers

Beyond $\aleph_0$: Hierarchies of Infinity

We've seen that $\aleph_0 = |\mathbb{N}|$ is the smallest infinite cardinality. But are there *larger* infinities beyond $\aleph_0$?

> Cantor showed that there are *infinitely many* infinite cardinal numbers, forming an endless hierarchy of ever-growing infinities.

Two key hierarchies help us organize these:

- The **ב (beth) numbers** — built by repeatedly taking powersets
- The **א (aleph) numbers** — indexing all infinite cardinals

Beth Numbers: The Powerset Hierarchy

Definition 61: The *beth numbers* are defined recursively using the powerset operation:

- $\beth_0 = \aleph_0 = |\mathbb{N}|$ (the first infinite cardinal)
- $\beth_{n+1} = 2^{\beth_n} = |\mathcal{P}(\beth_n)|$ (take the powerset)
- For limit ordinals λ: $\beth_\lambda = \sup\{\beth_\alpha \mid \alpha < \lambda\}$

Key insight: Each beth number is the cardinality of the powerset of the previous one.

This gives us a concrete, constructive hierarchy of infinities.

Beth Hierarchy: Examples

Example: The first few beth numbers:

- $\beth_0 = \aleph_0 = |\mathbb{N}|$ — countable infinity
- $\beth_1 = 2^{\beth_0} = 2^{\aleph_0} = |\mathcal{P}(\mathbb{N})| = |\mathbb{R}| = \mathfrak{c}$ — the *continuum*
- $\beth_2 = 2^{\beth_1} = 2^{\mathfrak{c}} = |\mathcal{P}(\mathbb{R})|$ — cardinality of the set of all functions $\mathbb{R} \to \mathbb{R}$
- $\beth_3 = 2^{\beth_2} = |\mathcal{P}(\mathcal{P}(\mathbb{R}))|$ — even larger!

Why this matters: We know from Cantor's theorem that $\beth_0 < \beth_1 < \beth_2 < \beth_3 < \ldots$ is a strictly increasing sequence. Each step is provably larger than the previous one!

Aleph Numbers: Indexing All Infinite Cardinals

Definition 62: The *aleph numbers* enumerate **all** infinite cardinal numbers in increasing order:

- $\aleph_0 = |\mathbb{N}|$ — the smallest infinite cardinal
- $\aleph_1$ — the *next* infinite cardinal after $\aleph_0$ (the smallest uncountable cardinal)
- $\aleph_2$ — the next infinite cardinal after $\aleph_1$
- In general: $\aleph_{\alpha+1}$ is the smallest cardinal larger than $\aleph_\alpha$
  - For limit ordinal λ: $\aleph_\lambda = \sup\{\aleph_\alpha \mid \alpha < \lambda\}$ (limit case)

Warning: Unlike beth numbers (defined by powerset), aleph numbers are defined by *order* — each is simply the "next" infinite cardinal. This makes them more abstract!

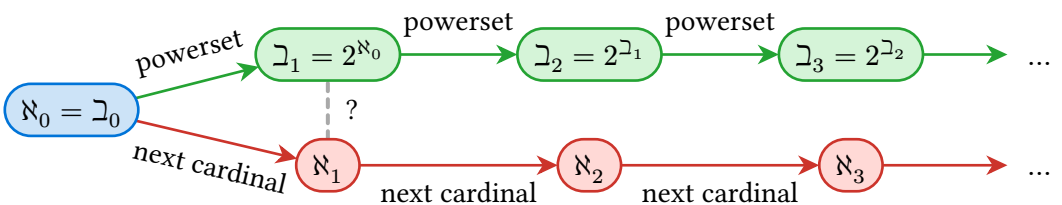
Aleph vs Beth: Two Different Hierarchies

Beth hierarchy (concrete):

- Start: $\beth_0 = \aleph_0$
- Rule: $\beth_{n+1} = 2^{\beth_n}$
- Built by taking powersets
- We know exactly what each $\beth_n$ is

Aleph hierarchy (ordinal):

- Start: $\aleph_0 = \beth_0$
- Rule: $\aleph_{n+1}$ is the *next* cardinal
- Built by well-ordering
- We know the *order* but not exact values



Both hierarchies start at $\aleph_0 = \beth_0$, but then *diverge*. The relationship between them is one of the deepest questions in set theory! **The key question**: is $\aleph_1 = \beth_1$?

The Continuum Hypothesis (CH)

Definition 63: The *Continuum Hypothesis* (CH) states that:

$$\aleph_1 = \beth_1 = 2^{\aleph_0} = \mathfrak{c}$$

In other words: there is *no* infinite cardinal strictly between $\aleph_0$ and $2^{\aleph_0}$.

Example: If CH is true:

- The continuum $\mathbb{R}$ has cardinality $\aleph_1$ (the first uncountable cardinal)
- Every uncountable subset of $\mathbb{R}$ has the same cardinality as $\mathbb{R}$
- The hierarchies align *at the first step*: $\aleph_1 = \beth_1$

Cantor's dream: He believed CH was true and spent years trying to prove it.

But the truth is far stranger...

The Generalized Continuum Hypothesis (GCH)

Definition 64 (GCH): For every infinite cardinal κ:

$$2^\kappa = \kappa^+$$

where κ^+ denotes the next cardinal after κ.

This means: $\aleph_n = \beth_n$ for all $n \in \mathbb{N}$ (and beyond).

GCH implies: The two hierarchies completely coincide — there's only one natural hierarchy of infinite cardinals, and powerset operation always gives you the very next cardinal.

Cohen's Independence Result

Theorem 17 (Cohen, 1963): The Continuum Hypothesis is *independent* of ZFC (Zermelo–Fraenkel set theory with Choice):

- We cannot *prove* CH from ZFC
- We cannot *disprove* CH from ZFC
- Both "ZFC + CH" and "ZFC + ¬CH" are consistent (if ZFC is)

What this means: Using *forcing*, Cohen showed that $\aleph_1$ can be placed *anywhere* relative to $\beth_1 = 2^{\aleph_0}$!

We can have:

- $\aleph_1 = \beth_1$ (CH is true)
- $\aleph_1 < \beth_1$ (there are cardinals between $\aleph_0$ and $2^{\aleph_0}$)
- Even: $\aleph_{100} = \beth_1$ (the continuum can be made arbitrarily large!)

The Freedom of Infinity

Key insight: The position of $2^{\aleph_0}$ in the aleph hierarchy is *not determined* by the usual axioms of mathematics!

In models where CH holds:

- $\aleph_1 = \beth_1 = 2^{\aleph_0}$
- Hierarchies align nicely
- Every subset of $\mathbb{R}$ is either countable or has the same cardinality $\mathfrak{c}$

In models where CH fails ($\neg$CH holds):

- $\aleph_1 < 2^{\aleph_0}$
- There exist "intermediate" infinities
- $\mathbb{R}$ can have very large cardinality
- The gap can be arbitrarily wide!

Historical note: This was the first major independence result in set theory, opening a new era where we study *which* mathematical universes are possible, not just what's true in one fixed universe.

Summary: Large Cardinal Numbers

Two infinite hierarchies:

- **Beth numbers:** $\beth_0 = \aleph_0$, $\beth_{n+1} = 2^{\beth_n}$ — concrete, built by powerset
- **Aleph numbers:** $\aleph_0$ smallest, $\aleph_{\alpha+1}$ next cardinal — abstract enumeration by order
- Both start at $\aleph_0 = \beth_0$ but then diverge — relationship unknown!
- **Continuum Hypothesis (CH):** Does $\aleph_1 = \beth_1 = 2^{\aleph_0}$?

Cohen (1963): CH is *independent* of ZFC — neither provable nor disprovable! Using forcing, $\aleph_1$ can be placed anywhere relative to $\beth_1$. The "size" of $\mathbb{R}$ is not determined by standard axioms.

Deep insight: Mathematics doesn't have a *unique* "reality" — different axiom systems create different mathematical universes, each equally valid!

Closures of Relations

Closures of Relations

Definition 65: The *closure* of a relation $R \subseteq M^2$ with respect to a property P is the smallest relation containing R that satisfies property P.

- *Reflexive closure*: $r(R) = R \cup I_M$ (smallest reflexive relation containing R)
- *Symmetric closure*: $s(R) = R \cup R^{-1}$ (smallest symmetric relation containing R)
- *Transitive closure*: $t(R)$ is the smallest transitive relation containing R

Key insight: closure operations *add the minimum* number of pairs needed to achieve the desired property, while preserving all existing pairs in the original relation.

Reflexive Closure

Definition 66: The *reflexive closure* $r(R)$ of a relation $R \subseteq M^2$ is defined as:

$$r(R) = R \cup I_M = R \cup \{\langle x, x\rangle \mid x \in M\}$$

Example: Let $M = \{1, 2, 3\}$ and $R = \{\langle 1, 2\rangle, \langle 2, 2\rangle, \langle 2, 3\rangle\}$.

The identity relation is $I_M = \{\langle 1, 1\rangle, \langle 2, 2\rangle, \langle 3, 3\rangle\}$.

The reflexive closure is:

$$r(R) = R \cup I_M = \{\langle 1, 1\rangle, \langle 1, 2\rangle, \langle 2, 2\rangle, \langle 2, 3\rangle, \langle 3, 3\rangle\}$$

Symmetric Closure

Definition 67: The *symmetric closure* $s(R)$ of a relation $R \subseteq M^2$ is defined as:

$$s(R) = R \cup R^{-1} = R \cup \{\langle b, a\rangle \mid \langle a, b\rangle \in R\}$$

Example: Let $M = \{1, 2, 3\}$ and $R = \{\langle 1, 2\rangle, \langle 2, 3\rangle, \langle 1, 3\rangle\}$.

The converse relation is:

$$R^{-1} = \{\langle 2, 1\rangle, \langle 3, 2\rangle, \langle 3, 1\rangle\}$$

The symmetric closure is:

$$s(R) = R \cup R^{-1} = \{\langle 1, 2\rangle, \langle 2, 3\rangle, \langle 1, 3\rangle, \langle 2, 1\rangle, \langle 3, 2\rangle, \langle 3, 1\rangle\}$$

Transitive Closure

Definition 68: The *transitive closure* $t(R)$ of a relation $R \subseteq M^2$ is the smallest transitive relation containing R.

Theorem 18: The transitive closure can be computed as:

$$t(R) = \bigcup_{n=1}^{\infty} R^n \quad \text{where } R^n = \underbrace{R \circ R \circ \ldots \circ R}_{n \text{ times}}$$

For finite sets with $|M| = k$, we have $t(R) = R^1 \cup R^2 \cup \ldots \cup R^k$.

Proof: Since M is finite, any path of length greater than $|M|$ must repeat vertices, so we only need to consider paths of length at most $|M|$. $\square$

Transitive Closure [2]

Example (Step-by-step transitive closure computation): Let $M = \{1, 2, 3\}$ and $R = \{\langle 1, 2\rangle, \langle 2, 3\rangle\}$.

Step	Description	Result
Step 1:	Compute $R^1 = R$.	$R^1 = \{\langle 1, 2\rangle, \langle 2, 3\rangle\}$
Step 2:	Compute $R^2 = R \circ R$.	$R^2 = \{\langle 1, 3\rangle\}$
	For $\langle a, c\rangle \in R^2$, we need $\exists b : \langle a, b\rangle \in R \wedge \langle b, c\rangle \in R$. • $\langle 1, 3\rangle \in R^2$ since $\langle 1, 2\rangle \in R$ and $\langle 2, 3\rangle \in R$	
Step 3:	Compute $R^3 = R^2 \circ R$.	$R^3 = \emptyset$
	For $\langle a, c\rangle \in R^3$, we need $\exists b : \langle a, b\rangle \in R^2 \wedge \langle b, c\rangle \in R$. • No such pairs exist.	
Step 4:	Form the transitive closure: $t(R) = R^1 \cup R^2 \cup R^3$.	$t(R) = \{\langle 1, 2\rangle, \langle 2, 3\rangle, \langle 1, 3\rangle\}$

Combined Closures

Definition 69: Closure operations can be combined to create relations with multiple properties:

- *Reflexive-symmetric closure*: $rs(R) = sr(R) = r(R) \cup s(R) \cup I_M$
- *Reflexive-transitive closure*: $rt(R) = tr(R) = t(R) \cup I_M$
- *Equivalence closure*: $rst(R) = tsr(R)$ (reflexive, symmetric, and transitive)

Theorem 19 (Commutativity of closure operations):

- Reflexive and symmetric closures commute: $rs(R) = sr(R)$
- Reflexive and transitive closures commute: $rt(R) = tr(R)$
- All three closures commute when applied together

Reflexive-Symmetric Closure

Example (Reflexive-symmetric closure): Let $M = \{1, 2, 3\}$ and $R = \{\langle 1, 2\rangle, \langle 2, 3\rangle\}$.

Method 1: Apply reflexive closure first, then symmetric:

$$\begin{aligned} r(R) &= R \cup I_M \\ &= \{\langle 1, 1\rangle, \langle 1, 2\rangle, \langle 2, 2\rangle, \langle 2, 3\rangle, \langle 3, 3\rangle\} \\ &\quad /\!/\ sr(R) = r(R) \cup r(R)^{-1} \\ &= \{\langle 1, 1\rangle, \langle 1, 2\rangle, \langle 2, 2\rangle, \langle 2, 3\rangle, \langle 3, 3\rangle, \langle 2, 1\rangle, \langle 3, 2\rangle\} \end{aligned}$$

Method 2: Apply symmetric closure first, then reflexive:

$$\begin{aligned} s(R) &= R \cup R^{-1} \\ &= \{\langle 1, 2\rangle, \langle 2, 3\rangle, \langle 2, 1\rangle, \langle 3, 2\rangle\} \\ &\quad /\!/\ rs(R) = s(R) \cup I_M \\ &= \{\langle 1, 1\rangle, \langle 1, 2\rangle, \langle 2, 1\rangle, \langle 2, 2\rangle, \langle 2, 3\rangle, \langle 3, 2\rangle, \langle 3, 3\rangle\} \end{aligned}$$

Both methods yield the same result, confirming *commutativity*.

Reflexive-Transitive Closure

Example (Reflexive-transitive closure (Kleene star)): Let $M = \{a, b, c\}$ and $R = \{\langle a, b\rangle, \langle b, c\rangle\}$.

First, compute the transitive closure:

$$t(R) = R \cup R^2 = \{\langle a, b\rangle, \langle b, c\rangle, \langle a, c\rangle\}$$

Then add reflexivity:

$$rt(R) = t(R) \cup I_M = \{\langle a, a\rangle, \langle a, b\rangle, \langle a, c\rangle, \langle b, b\rangle, \langle b, c\rangle, \langle c, c\rangle\}$$

This is equivalent to the *reflexive-transitive closure*, often denoted R^* (Kleene star).

Equivalence Closure

Example: Let $M = \{1, 2, 3, 4\}$ and $R = \{\langle 1, 2\rangle, \langle 3, 4\rangle\}$.

Step 1: Make it reflexive:

$$r(R) = R \cup I_M = \{\langle 1, 1\rangle, \langle 1, 2\rangle, \langle 2, 2\rangle, \langle 3, 3\rangle, \langle 3, 4\rangle, \langle 4, 4\rangle\}$$

Step 2: Make it symmetric:

$$\begin{aligned} sr(R) &= r(R) \cup r(R)^{-1} \\ &= \{\langle 1, 1\rangle, \langle 1, 2\rangle, \langle 2, 1\rangle, \langle 2, 2\rangle, \langle 3, 3\rangle, \langle 3, 4\rangle, \langle 4, 3\rangle, \langle 4, 4\rangle\} \end{aligned}$$

Step 3: Make it transitive:

- Since $\langle 1, 2\rangle, \langle 2, 1\rangle \in sr(R)$, transitivity requires $\langle 1, 1\rangle$ (already present).
- Since $\langle 3, 4\rangle, \langle 4, 3\rangle \in sr(R)$, transitivity requires $\langle 3, 3\rangle$ (already present).

$$tsr(R) = sr(R) \quad \text{(no new pairs needed)}$$

The equivalence closure partitions M into equivalence classes $\{1, 2\}$ and $\{3, 4\}$.

Equivalence Closure [2]

Example: Let $M = \{a, b, c, d, e\}$ and $R = \{\langle a, b\rangle, \langle b, c\rangle, \langle d, e\rangle\}$.

Reflexive closure:

$$r(R) = R \cup \{\langle a, a\rangle, \langle b, b\rangle, \langle c, c\rangle, \langle d, d\rangle, \langle e, e\rangle\}$$

Symmetric closure:

$$sr(R) = r(R) \cup \{\langle b, a\rangle, \langle c, b\rangle, \langle e, d\rangle\}$$

Transitive closure: We need to add pairs to ensure transitivity:

- From $\langle a, b\rangle, \langle b, c\rangle$: add $\langle a, c\rangle$
- From $\langle c, b\rangle, \langle b, a\rangle$: add $\langle c, a\rangle$

$$tsr(R) = sr(R) \cup \{\langle a, c\rangle, \langle c, a\rangle\}$$

The final equivalence relation has equivalence classes $\{a, b, c\}$ and $\{d, e\}$.

Warshall's Algorithm for Transitive Closure

Definition 70: *Warshall's algorithm* computes the transitive closure of a relation using dynamic programming with time complexity $O(n^3)$.

Given an $n \times n$ matrix $M = [\![R]\!]$ representing relation R:

```
M = matrix(R)
for k = 1 to n:
    for i = 1 to n:
        for j = 1 to n:
            M[i,j] := M[i,j] OR (M[i,k] AND M[k,j])
```

Example (Warshall's algorithm step-by-step): Let $X = \{1, 2, 3, 4\}$ and relation R with matrix $[\![R]\!]$:

$$M^{(0)} = [\![R]\!] = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

Warshall's Algorithm for Transitive Closure [2]

Iteration $k = 1$: Consider paths through vertex 1.

$$M^{(1)} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$

(Added $\langle 4, 2 \rangle$: path $4 \to 1 \to 2$)

Iteration $k = 2$: Consider paths through vertex 2.

$$M^{(2)} = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

(Added $\langle 1, 3 \rangle$ and $\langle 4, 3 \rangle$)

Warshall's Algorithm for Transitive Closure [3]

Iteration $k = 3$: Consider paths through vertex 3.

$$M^{(3)} = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

(Added $\langle 1, 4 \rangle$, $\langle 2, 4 \rangle$, and $\langle 4, 4 \rangle$)

Iteration $k = 4$: Consider paths through vertex 4.

$$M^{(4)} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

(The relation becomes universal since there's a cycle)

Warshall's Algorithm for Transitive Closure [4]

Example (Reachability in graphs):

Consider a social network where R represents "follows" relationships:

$$R = \{\langle A, B\rangle, \langle B, C\rangle, \langle C, D\rangle, \langle A, E\rangle\}$$

Using Warshall's algorithm, we can determine *indirect influence*:

- A can influence C through B
- A can influence D through B and C
- The transitive closure shows all possible influence paths

This is crucial for analyzing *information propagation* in social networks, *dependency resolution* in software systems, and *route planning* in transportation networks.

Properties of Closures

Theorem 20: For any relation $R \subseteq M^2$:

1. *Idempotency*: $r(r(R)) = r(R)$, $s(s(R)) = s(R)$, $t(t(R)) = t(R)$
2. *Monotonicity*: If $R_1 \subseteq R_2$, then $r(R_1) \subseteq r(R_2)$, *etc.*
3. *Extensivity*: $R \subseteq r(R)$, $R \subseteq s(R)$, $R \subseteq t(R)$
4. *Distributivity over union*: $r(R_1 \cup R_2) = r(R_1) \cup r(R_2)$, *etc.*

Examples of Closures

Example (Closure of the empty relation):

Let $M = \{a, b, c\}$ and $R = \emptyset$.

- $r(\emptyset) = \emptyset \cup I_M = I_M = \{\langle a, a\rangle, \langle b, b\rangle, \langle c, c\rangle\}$
- $s(\emptyset) = \emptyset \cup \emptyset^{-1} = \emptyset$
- $t(\emptyset) = \emptyset$ (since $\emptyset^n = \emptyset$ for all $n \geq 1$)

The reflexive closure of the *empty* relation is the *identity* relation.

Examples of Closures [2]

Example (Closure of the universal relation):

Let $M = \{1, 2\}$ and $R = M \times M = \{\langle 1, 1\rangle, \langle 1, 2\rangle, \langle 2, 1\rangle, \langle 2, 2\rangle\}$.

- $r(R) = R \cup I_M = R$ (since $I_M \subseteq R$)
- $s(R) = R \cup R^{-1} = R$ (since $R = R^{-1}$ for universal relation)
- $t(R) = R$ (universal relation is already transitive)

The universal relation is its own closure under all three operations.

Examples of Closures [3]

Example (Non-commutativity with other operations):

Let $M = \{1, 2, 3\}$, $R_1 = \{\langle 1, 2\rangle\}$, and $R_2 = \{\langle 2, 3\rangle\}$.

Consider $t(R_1 \cup R_2)$ vs $t(R_1) \cup t(R_2)$:

- $R_1 \cup R_2 = \{\langle 1, 2\rangle, \langle 2, 3\rangle\}$
- $t(R_1 \cup R_2) = \{\langle 1, 2\rangle, \langle 2, 3\rangle, \langle 1, 3\rangle\}$
- $t(R_1) = \{\langle 1, 2\rangle\}$
- $t(R_2) = \{\langle 2, 3\rangle\}$
- $t(R_1) \cup t(R_2) = \{\langle 1, 2\rangle, \langle 2, 3\rangle\}$

Since $\langle 1, 3\rangle \in t(R_1 \cup R_2)$ but $\langle 1, 3\rangle \notin t(R_1) \cup t(R_2)$, we have:

$$t(R_1 \cup R_2) \neq t(R_1) \cup t(R_2)$$

However: $t(R_1) \cup t(R_2) \subseteq t(R_1 \cup R_2)$ always holds.

Examples of Closures [4]

Example (Computing equivalence classes from closure):

Let $M = \{1, 2, 3, 4, 5\}$ and $R = \{\langle 1, 3\rangle, \langle 2, 4\rangle, \langle 4, 5\rangle\}$.

The equivalence closure gives us:

$$\begin{aligned}\text{equiv}(R) &= rst(R) \\ &= \{\langle 1, 1\rangle, \langle 1, 3\rangle, \langle 2, 2\rangle, \langle 2, 4\rangle, \langle 2, 5\rangle, \langle 3, 1\rangle, \\ &\qquad \langle 3, 3\rangle, \langle 4, 2\rangle, \langle 4, 4\rangle, \langle 4, 5\rangle, \langle 5, 2\rangle, \langle 5, 4\rangle, \langle 5, 5\rangle\}\end{aligned}$$

The equivalence classes are:

- $[1]_R = \{1, 3\}$
- $[2]_R = \{2, 4, 5\}$

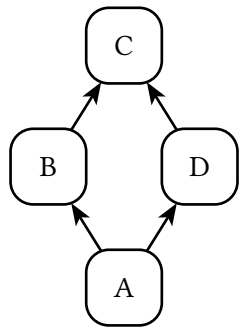
This partitions M into $\{\{1, 3\}, \{2, 4, 5\}\}$.

Examples of Closures [5]

Example: Consider a directed acyclic graph (DAG) where R represents *"depends on"* relationships:

$$R = \{\langle A, B\rangle, \langle B, C\rangle, \langle A, D\rangle, \langle D, C\rangle\}$$

For example, component A depends on B and D, which both depend on C.



The transitive closure reveals all *indirect* dependencies:

$$t(R) = R \cup \{\langle A, C\rangle\}$$

This shows that component A *transitively depends* on C through two paths:

- $A \to B \to C$
- $A \to D \to C$

In software build systems, this helps determine the complete dependency tree.

Applications of Relation Closures

Example ***(Graph Reachability)***: In directed graphs, transitive closure determines reachability:

- **Social networks:** Finding mutual connections and influence propagation
- **Transportation:** Computing all possible routes between cities
- **Citation networks:** Tracking intellectual dependencies in research
- **Web crawling:** Discovering linked pages and site connectivity

Example ***(Program Analysis)***: Closure operations enable sophisticated program optimization:

- **Data flow analysis:** Transitive closure computes variable dependencies
- **Call graph construction:** Finding all possible function invocations
- **Alias analysis:** Determining which pointers may reference the same memory
- **Taint analysis:** Tracking information flow and potential vulnerabilities

Example ***(Database Systems)***: Relational databases extensively use closure computations:

- **Recursive queries:** Computing organizational hierarchies and bill-of-materials
- **Join optimization:** Finding efficient query execution plans
- **Integrity constraints:** Ensuring referential consistency across tables
- **Data lineage:** Tracking data provenance and transformation chains

Complexity of Closure Computations

Theorem 21: For a relation R on n elements, the time complexities are:

- **Reflexive closure:** $O(n)$ time
- **Symmetric closure:** $O(|R|)$ time
- **Transitive closure:** $O(n^3)$ time (Floyd-Warshall algorithm)

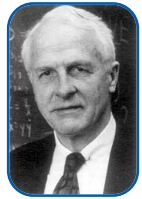
Note: The transitive closure bottleneck drives many algorithmic design decisions in practice.

Lattices

"Order is the shape upon which beauty depends."

— Pearl S. Buck

Alfred Tarski



Garrett Birkhoff

Dana Scott

Emmy Noether

Marshall Stone

Upper and Lower Bounds

Definition 71: In a poset $\langle S, \leq \rangle$, an element $u \in S$ is called an *upper bound* of a subset $C \subseteq S$ if it is greater than or equal to every element in C, *i.e.*, for all $x \in C$, $x \leq u$.

Definition 72: In a poset $\langle S, \leq \rangle$, an element $l \in S$ is called a *lower bound* of a subset $C \subseteq S$ if it is less than or equal to every element in C, *i.e.*, for all $x \in C$, $l \leq x$.

Example: In $\langle \mathbb{R}, \leq \rangle$ for interval $C = (0; 1)$:

- **Lower bounds:** every $x \leq 0$ (including $-\infty, -1, 0$)
- **Upper bounds:** every $x \geq 1$ (including $1, 2, +\infty$)
- **No** greatest lower bound or least upper bound *in* C (since $(0; 1)$ is open)

Examples of Bounds

Example: In $\langle \mathcal{P}(\{1,2,3\}), \subseteq \rangle$ for $C = \{\{1,2\},\{1,3\}\}$:

- **Lower bounds:** $\emptyset$, $\{1\}$ (subsets of both sets in C)
- **Upper bounds:** $\{1,2,3\}$ (supersets of both sets in C)
- **Greatest lower bound:** $\{1\} = \{1,2\} \cap \{1,3\}$
- **Least upper bound:** $\{1,2,3\} = \{1,2\} \cup \{1,3\}$

Example: In divisibility poset for $C = \{4,6\}$:

- **Upper bounds:** multiples of $\mathrm{lcm}(4,6) = 12$, *i.e.*, $\{12, 24, 36, \ldots\}$
- **Lower bounds:** common divisors, *i.e.*, $\{1,2\}$
- **Least upper bound:** $12 = \mathrm{lcm}(4,6)$
- **Greatest lower bound:** $2 = \gcd(4,6)$

Suprema and Infima

Definition 73: In a poset $\langle S, \leq \rangle$, the *supremum* (or *join*) of a subset $C \subseteq S$, denoted $\sup(C)$ or $\bigvee C$, is the *least upper bound* of C, *i.e.*, an upper bound $u \in S$ s.t. for any other upper bound $v \in S$, $u \leq v$.

Note: If it exists, the least upper bound is *unique*.

Definition 74: In a poset $\langle S, \leq \rangle$, the *infimum* (or *meet*) of a subset $C \subseteq S$, denoted $\inf(C)$ or $\bigwedge C$, is the *greatest lower bound* of C, *i.e.*, a lower bound $l \in S$ s.t. for any other lower bound $m \in S$, $m \leq l$.

Note: If it exists, the greatest lower bound is *unique*.

Example: $\langle \mathbb{R}, \leq \rangle$:

- For finite subsets, $\sup(C) = \max(C)$ and $\inf(C) = \min(C)$.
- For infinite subsets: $\sup((0;1)) = 1$ and $\inf((0;1)) = 0$ (even though $0, 1 \notin (0;1)$)

Examples of Suprema and Infima

Example: $\langle \mathcal{P}(A), \subseteq \rangle$:

- **Join:** $\sup(\mathcal{C}) = \bigcup_{X \in \mathcal{C}} X$ (union of all sets)
- **Meet:** $\inf(\mathcal{C}) = \bigcap_{X \in \mathcal{C}} X$ (intersection of all sets)
- $\sup\{\{1,2\},\{2,3\},\{3,4\}\} = \{1,2,3,4\}$
- $\inf\{\{1,2,3\},\{2,3,4\},\{3,4,5\}\} = \{3\}$

Example: Divisibility on $\mathbb{N}^+$:

- **Join:** $\sup\{a,b\} = \mathrm{lcm}(a,b)$ (least common multiple)
- **Meet:** $\inf\{a,b\} = \gcd(a,b)$ (greatest common divisor)
- $\sup\{6,10\} = 30$
- $\inf\{6,10\} = 2$

Lattices

Definition 75: A poset $\langle S, \leq \rangle$ where every non-empty finite subset $C \subseteq S$ has a join (supremum) is called an *upper semilattice* (or *join-semilattice*) and denoted $\langle S, \vee \rangle$.

Definition 76: A poset $\langle S, \leq \rangle$ where every non-empty finite subset $C \subseteq S$ has a meet (infimum) is called a *lower semilattice* (or *meet-semilattice*) and denoted $\langle S, \wedge \rangle$.

Definition 77: A poset $\langle S, \leq \rangle$ that is both an upper semilattice and a lower semilattice, *i.e.*, every non-empty finite subset has both a join and a meet, is called a *lattice*, denoted $(S, \vee, \wedge)$.

Why Lattices?

Why study lattices? Whenever you have:

- Elements that can be *compared* (ordered)
- Ways to *combine* elements (join, meet)
- Consistent behavior under combination

...you likely have a lattice! This structure appears in programming languages, databases, security systems, logic circuits, and many other areas of computer science and mathematics.

Properties of Lattices

Definition 78: A lattice is *bounded* if it has a greatest element $\top$ and a least element $\bot$.

Definition 79: A lattice is *distributive* if $x \wedge (y \vee z) = (x \wedge y) \vee (x \wedge z)$ (and dually).

Definition 80: A lattice is *modular* if $x \leq z$ implies $x \vee (y \wedge z) = (x \vee y) \wedge z$.

Note: Distributive $\Rightarrow$ modular.

Example (Powerset Lattice): $\langle \mathcal{P}(A), \subseteq \rangle$ is a bounded distributive lattice with $\vee = \cup$, $\wedge = \cap$, $\top = A$, $\bot = \emptyset$.

Why this matters: This is the foundation of set-based reasoning in:

- Database theory (relational algebra)
- Formal specification languages (Z, B-method)
- Model checking and verification

Examples of Lattices

Example (Divisibility Lattice): For positive integers, $a \leq b$ iff a divides b.

- **Join (LCM):** $6 \vee 10 = \mathrm{lcm}(6, 10) = 30$
- **Meet (GCD):** $6 \wedge 10 = \gcd(6, 10) = 2$
- **Bottom element:** 1 (divides everything)
- **No top element:** No integer is divisible by all others

Applications:

- Number theory and cryptography (RSA key generation)
- Computer algebra systems (polynomial GCD algorithms)
- Scheduling problems (finding common time periods)

Examples of Lattices [2]

Example (File System Permissions): Unix file permissions form a lattice under inclusion:

- Elements: Sets of permissions like $\{r, w, x\}$, $\{r, x\}$, $\{w\}$, *etc.*
- Order: $P_1 \leq P_2$ if $P_1 \subseteq P_2$ (fewer permissions $\Rightarrow$ more restrictive)
- Join: Union of permissions (less restrictive)
  - For example: $\{\text{read}\} \vee \{\text{execute}\} = \{\text{read}, \text{execute}\}$
- Meet: Intersection of permissions (more restrictive)
  - For example: $\{\text{read}, \text{write}\} \wedge \{\text{write}, \text{execute}\} = \{\text{write}\}$

Examples of Lattices [3]

Example (Partition Lattice): All partitions of a set S, ordered by refinement.

- $\pi_1 \leq \pi_2$ if π_1 is a refinement of π_2 (smaller blocks)
- Join: Finest common coarsening
  - For example: $(12 \mid 3) \vee (1 \mid 23) = (123)$
- Meet: Coarsest common refinement
  - For example: $(12 \mid 3) \wedge (1 \mid 23) = (1 \mid 2 \mid 3)$
- Applications: Clustering, database normalization

Lattices aren't just abstract algebra — they appear everywhere in computer science and mathematics.

The *join* and *meet* operations capture fundamental patterns of *combination* and *interaction*.

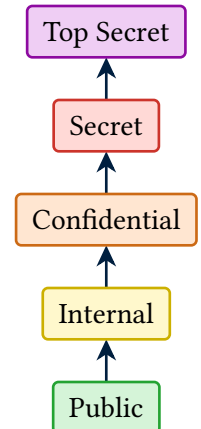
Why Lattices Matter: Information Security Levels

Example: In computer security, information has *classification levels* forming a lattice:

- Elements: $\{\text{Public}, \text{Internal}, \text{Confidential}, \text{Secret}, \text{Top Secret}\}$
- Order: $\text{Public} \leq \text{Internal} \leq \text{Confidential} \leq \text{Secret} \leq \text{Top Secret}$
- Join ($\vee$): Higher classification needed to combine information
- Meet ($\wedge$): Lower classification that both pieces can be declassified to

For instance:

- $\text{Internal} \vee \text{Confidential} = \text{Confidential}$ (combination needs higher level)
- $\text{Secret} \wedge \text{Confidential} = \text{Confidential}$ (both can be declassified to this level)



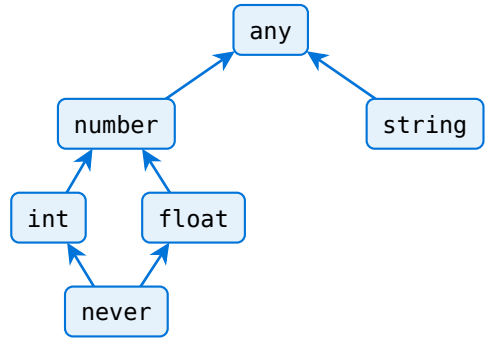
Why Lattices Matter: Type Systems

In programming language theory, *types* form lattices.

```
let x: int = 42;
let y: string = "hello";
let z = condition ? x : y;
// z has type: int ∨ string = any
```

Subtyping Lattice:

- Order: `int` $\sqsubseteq$ `number` $\sqsubseteq$ `any`, `string` $\sqsubseteq$ `any`
- Join ($\vee$): Most general common supertype (union type)
  - For example: `int` $\vee$ `string` $=$ `any`
- Meet ($\wedge$): Most specific common subtype (intersection type)
  - Let $A =$ `{ name: string }`
  - Let $B =$ `{ age: int }`
  - Then $A \wedge B =$ `{ name: string, age: int }`



Type Lattice

- Arrows show subtype relation $\sqsubseteq$
- Join: Move up to common parent
- Meet: Move down to common child

Why Lattices Matter: Program Analysis

TODO

Control Flow Analysis:

- Elements: Sets of possible program states
- Order: Subset inclusion ($\subseteq$)
- Join: Union of possible states (at merge points)
- Meet: Intersection of guaranteed properties

Why Lattices Matter: Database Query Optimization

Example: *Query execution plans* form a lattice:

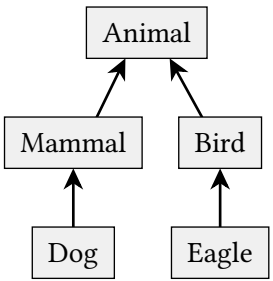
- Elements: Different ways to execute a query
- Order: "Plan A $\leq$ Plan B" if A is more efficient than B
- Join: Combine optimization strategies
- Meet: Find common optimizations

This structure helps database optimizers systematically explore the space of possible query plans.

Why Lattices Matter: Concept Hierarchies and Ontologies

Example: Knowledge representation uses *concept lattices*.

For example, consider a biological taxonomy:



- Elements: Biological concepts (*e.g.*, Animal, Mammal, Dog)
- Order: "Concept A $\leq$ Concept B" if A is a more specific type of B, *e.g.*, "Dog $\leq$ Mammal"
- Join: Most specific common ancestor, *e.g.*, "Mammal $\vee$ Bird $=$ Animal"
- Meet: Most general common descendant, *e.g.*, "Bird $\wedge$ Eagle $=$ Eagle"

Why Lattices Matter: Distributed Systems and Causality

Example: In distributed systems, *events* form a lattice *under causality*:

- Elements: System events with vector timestamps
- Order: "Event A $\leq$ Event B" if A causally precedes B
- Join: Latest information from both events
- Meet: Common causal history

This structure is crucial for:

- Consistent distributed databases
- Version control systems (Git DAG)
- Blockchain consensus algorithms

Why Lattices Matter: Logic and Boolean Reasoning

Example: *Propositional formulas* form lattices:

- Elements: Boolean formulas over variables
- Order: $\varphi \leq \psi$ if φ implies ψ (semantic entailment)
- Join: Disjunction ($\vee$) — weaker condition
- Meet: Conjunction ($\wedge$) — stronger condition

Special case: *Boolean algebra* (`true`, `false`, $\vee$, $\wedge$, $\neg$) used in:

- Digital circuit design
- Database query languages (SQL WHERE clauses)
- Search engines (Boolean search)

Well Orders

Well-Ordered Sets

Definition 81: A poset $\langle M, \leq \rangle$ is *well-ordered* if every non-empty subset $S \subseteq M$ has a *least element*.

Formally: $\forall S \subseteq M. (S \neq \emptyset) \to (\exists m \in S. \forall x \in S. m \leq x)$

Note: A well-ordered set is automatically a *total order* (linear order) since comparability follows from the well-ordering property.

Example: The natural numbers $\mathbb{N} = \{0, 1, 2, 3, \ldots\}$ with the usual order $\leq$ form a well-ordered set:

- Any non-empty subset has a smallest element.
- For instance, the subset $\{5, 17, 23, 100\}$ has least element 5.
- Even infinite subsets like $\{2, 4, 6, 8, \ldots\}$ (even numbers) have a least element (2).

Example: The integers $\mathbb{Z}$ with the usual order are *not* well-ordered:

- The subset $\{-1, -2, -3, \ldots\}$ (negative integers) has no least element.
- Any infinite descending sequence has no minimum.

Examples of Well-Ordered Sets

Example: *Lexicographic order* on finite strings over an alphabet is well-ordered:

- Given any non-empty set of strings, there is always a lexicographically smallest one.
- For example, in $\{\text{cat}, \text{dog}, \text{apple}, \text{zebra}\}$, the least element is "apple".

Well-Ordered Induction

TODO

Well-Founded Relations

Definition 82: A relation $R \subseteq M^2$ is *well-founded* if every non-empty subset $S \subseteq M$ has at least one *minimal element* with respect to R.

Formally: $\forall S \subseteq M.\, (S \neq \emptyset) \to (\exists m \in S.\, \forall x \in S.\, x \not R\, m)$

Note: *Well-founded* $\neq$ *well-ordered*:

- Well-ordered requires a *least* element (unique minimum)
- Well-founded only requires *minimal* elements (no element below them)
- Every well-ordered set is well-founded, but not vice versa

Examples of Well-Founded Relations

Example: Consider the *proper subset* relation $\subset$ on finite sets. Let $M = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$.

Well-founded: ✓ Every subset of M has minimal elements.

- *Example:* The subset $\{\{a\}, \{b\}, \{a, b\}\}$ has minimal elements $\{a\}$ and $\{b\}$
  - Neither $\{a\} \subset \{b\}$ nor $\{b\} \subset \{a\}$ (they're incomparable)
  - Both are minimal since no set in the subset is a proper subset of them

Well-ordered: ✗ Some subsets lack a unique least element.

- *Same example:* $\{\{a\}, \{b\}, \{a, b\}\}$ has no "$\subset$-least" element
  - For "$\subset$-least", we'd need a set L such that $L \subset X$ for all other X
  - But $\{a\} \not\subset \{b\}$ and $\{b\} \not\subset \{a\}$, so neither can be least
  - No single set is a proper subset of all others in this collection

Key insight: Well-founded $\neq$ well-ordered:

- *Multiple minimal* elements are allowed in well-founded relations.
- Well-ordered relations require a *unique* least element in every subset.

Examples of Well-Founded Relations [2]

Example: The *divisibility* relation $\langle \mathbb{N}^+, | \rangle$ is well-founded:

- Every non-empty subset has minimal elements (numbers that divide no others in the subset)
- For example, in $\{6, 12, 18, 4, 8\}$:
  - 4 is minimal because no other number in the set divides 4
  - 6 is minimal because no other number in the set divides 6
  - Note: 6 divides both 12 and 18, but that doesn't affect minimality
- In $\{2, 4, 8, 16\}$: only 2 is minimal (it divides all others, but nothing else divides it)

Examples of Well-Founded Relations [3]

Example: Consider the relation *"properly contains"* ($\supset$) on finite sets.

Let $S = \{\{1\}, \{2\}, \{1, 2\}, \{1, 2, 3\}\}$.

Well-founded: ✓ Every subset has minimal elements.

- The subset $\{\{1\}, \{2\}, \{1, 2, 3\}\}$ has minimal elements $\{1\}$ and $\{2\}$
  - Neither $\{1\} \supset \{2\}$ nor $\{2\} \supset \{1\}$ (they're incomparable)
  - $\{1, 2, 3\} \supset \{1\}$ and $\{1, 2, 3\} \supset \{2\}$, but they remain minimal in this subset
- Any collection always has sets that contain no others in that collection

Examples of Well-Founded Relations [4]

Example: *Program termination analysis* uses well-founded relations:

- Define a measure that decreases with each recursive call
- If the measure forms a well-founded order, the program terminates
- Example: factorial function decreases argument from n to $n - 1$

Well-Founded Induction

TODO

Induced Strict Order

Definition 83: Given a partial order $\leq$, the *induced strict order* $<$ is defined as:

$$x < y \text{ iff } (x \leq y \text{ and } x \neq y)$$

Note: Given a poset $\langle S, \leq \rangle$, we can use $<$ to denote its associated strict order.

Note: $>$ is the converse of $<$, so we can write $b > a$ instead of $a < b$.

Note: *Hereinafter*, we will freely use $<$ and $>$ when given any poset $\langle S, \leq \rangle$.

Descending Chain Condition

Definition 84: A poset $\langle S, \leq \rangle$ is said to satisfy the *descending chain condition (DCC)* if no strict descending sequence $x_1 > x_2 > x_3 > \ldots$ of elements of S exists.

Equivalently, every weakly descending sequence $x_1 \geq x_2 \geq x_3 \geq \ldots$ eventually stabilizes.

Formally: $\forall (x_i)_{i \in \mathbb{N}} \in S^{\mathbb{N}}. \left(\forall i \in \mathbb{N}. x_i \geq x_{i+1}\right) \to \left(\exists N \in \mathbb{N}. \forall n \geq N. x_n = x_{n+1}\right)$

Note: The term "*sequence*" does not mean we must list all elements consequently or uniquely. It simply means we have a function from $\mathbb{N}$ to S. Elements can repeat, and we can skip elements in S. However, when we say "*descending sequence*," we mean that each new element is less than the previous one.

Example: The natural numbers $\langle \mathbb{N}, \leq \rangle$ satisfy DCC:

- Since natural numbers are bounded below by 0, infinite descent is impossible.
- Any (weakly) descending sequence $n_1 \geq n_2 \geq n_3 \geq \ldots$ must stabilize.
- Eventually, some $n_k = n_{k+1} = n_{k+2} = \ldots$

Well-Founded Posets

Definition 85: A poset $\langle S, \leq \rangle$ is *well-founded* if its associated strict order $<$ is a well-founded relation.

In other words, every non-empty subset of S has $<$-minimal elements.

DCC and Well-Foundedness

Theorem 22: For any poset $\langle S, \leq \rangle$, the following are *equivalent*:

1. $\langle S, \leq \rangle$ is *well-founded* (every non-empty subset has $<$-minimal elements)
2. $\langle S, \leq \rangle$ satisfies *DCC* (no infinite descending chains)

Note: Here, we again use "$<$" (and its converse "$>$") to denote the strict order induced by "$\leq$".

Proof *($\Longrightarrow$)*: **Well-founded implies DCC**

Suppose $\langle S, \leq \rangle$ is well-founded but there exists an infinite strict descending sequence $x_0 > x_1 > x_2 > \ldots$. Consider the set $T = \{x_0, x_1, x_2, \ldots\}$. Since $\langle S, \leq \rangle$ is well-founded, T must have a minimal element x_k for some k. But then $x_k > x_{k+1}$, contradicting the minimality of x_k. $\square$

Proof *($\Longleftarrow$)*: **DCC implies well-founded**

Suppose $\langle S, \leq \rangle$ satisfies DCC. Let $T \subseteq S$ be any non-empty subset. If T had no minimal elements, then for any $x_0 \in T$, there would exist $x_1 \in T$ with $x_0 > x_1$. Continuing this process, we could construct an infinite strict descending sequence $x_0 > x_1 > x_2 > \ldots$, contradicting DCC. $\square$

Ascending Chain Condition

Definition 86: A poset $\langle S, \leq \rangle$ satisfies the *ascending chain condition (ACC)* if no strict ascending sequence $x_1 < x_2 < x_3 < \ldots$ of elements of S exists.

Equivalently, every weakly ascending sequence $x_1 \leq x_2 \leq x_3 \leq \ldots$ eventually stabilizes.

Example: $\langle \mathbb{N}, \leq \rangle$ does *not* satisfy ACC since infinite ascending chains like $2 < 3 < 5 < 7 < \ldots$ exist.

Example: Consider the poset $\langle \mathcal{P}(\mathbb{N}), \subseteq \rangle$ of all subsets of natural numbers ordered by inclusion.

- **ACC fails:** ✗
  The ascending chain $\emptyset \subseteq \{1\} \subseteq \{1, 2\} \subseteq \{1, 2, 3\} \subseteq \ldots$ never stabilizes.
- **DCC fails:** ✗
  The descending chain $\mathbb{N} \supseteq (\mathbb{N} \setminus \{1\}) \supseteq (\mathbb{N} \setminus \{1, 2\}) \supseteq \ldots$ never stabilizes.

Noetherian Relations

Definition 87: A relation R is *Noetherian* (or *converse well-founded*, or *upwards well-founded*) if the converse relation R^{-1} is well-founded.

Equivalently, $R \subseteq M^2$ is Noetherian if every non-empty subset of M has R-maximal elements.

Note: This means no infinite ascending chains $x_0 \; R \; x_1 \; R \; x_2 \; R \; \ldots$ exist.

Example: The usual order $\leq$ on $\mathbb{N}$ is NOT Noetherian:

- We can construct infinite ascending chains like $1 < 2 < 3 < 4 < \ldots$
- However, its converse $\geq$ would be well-founded (DCC holds for $\leq$)

Noetherian Relations [2]

Example: In ring theory, a *Noetherian ring* has the property that every ascending chain of ideals stabilizes: $I_1 \subseteq I_2 \subseteq I_3 \subseteq \ldots$ eventually becomes constant.

Example: In rewriting systems and lambda calculus:

- A *reduction relation* $\rightarrow$ is Noetherian if all reduction sequences terminate
- For example, β-reduction in simply typed lambda calculus is Noetherian
- This guarantees that programs always terminate (no infinite loops)

Noetherian Relations and ACC

Theorem 23: For any poset $\langle S, \leq \rangle$, the following are equivalent:

- $\langle S, \leq \rangle$ is Noetherian (coverse well-founded)
- $\langle S, \leq \rangle$ satisfies ACC (no infinite ascending chains)

Proof: By definition, $\langle S, \leq \rangle$ is Noetherian iff its converse $\geq$ is well-founded. By the earlier theorem, this is equivalent to $\geq$ satisfying DCC, which is exactly the same as $\leq$ satisfying ACC. $\square$

Relationships Between Chain Conditions

Theorem 24: For a poset $\langle S, \leq \rangle$:

- **DCC** $\iff$ the relation $\leq$ is well-founded $\iff$ no infinite descending chains.
- **ACC** $\iff$ the *dual* relation $\geq$ is well-founded $\iff$ no infinite ascending chains.

Proof: The first equivalence (DCC $\iff$ well-founded) was proved earlier. The second equivalence for ACC follows by applying the same reasoning to the dual relation $\geq$. $\square$

Examples of ACC and DCC

Example: Consider the poset $\langle \mathcal{P}(\{1, 2, 3\}), \subseteq \rangle$ of subsets ordered by inclusion:

- **ACC holds:** ✓ Any ascending chain $A_1 \subseteq A_2 \subseteq A_3 \subseteq \ldots$ must stabilize since we can't keep adding elements indefinitely.
- **DCC holds:** ✓ Any descending chain $B_1 \supseteq B_2 \supseteq B_3 \supseteq \ldots$ must stabilize since we can't keep removing elements indefinitely.
- Both conditions hold because the set is finite.

Example: In the natural numbers $\langle \mathbb{N}, \leq \rangle$:

- **DCC holds:** ✓ Any sequence $n_1 \geq n_2 \geq n_3 \geq \ldots$ must stabilize (well-founded).
- **ACC fails:** ✗ The sequence $1 < 2 < 3 < 4 < \ldots$ never stabilizes.
- This shows that DCC and ACC are independent conditions.

Examples (Applications in algebra):

- **Noetherian rings:** Satisfy ACC for ideals (every ascending chain of ideals stabilizes).
- **Artinian rings:** Satisfy DCC for ideals (every descending chain of ideals stabilizes).
- **Principal ideal domains:** Both conditions hold, enabling algorithms like Euclidean division.

Connections and Applications

Theorem 25 (Well-ordering principle): Every well-ordered set admits *transfinite induction*: to prove $P(x)$ for all $x \in S$, it suffices to show: $\forall x \in S.\, (\forall y < x.\, P(y)) \to P(x)$

Example: *Mathematical induction* on $\mathbb{N}$ is a special case of transfinite induction using the well-ordering of natural numbers.

Examples (Computer science applications):

- **Termination analysis:** Prove programs terminate by finding well-founded measures.
- **Parsing algorithms:** Use well-founded recursion on parse tree depth.
- **Datalog evaluation:** Stratified negation ensures termination via well-founded semantics.
- **Model checking:** Well-founded relations ensure finite state exploration.

These concepts provide the mathematical foundation for reasoning about *termination*, *finiteness*, and *algorithmic complexity* in computer science and mathematics.

Summary: Well-Founded Relations and Chain Conditions

Concept	Characterization	Key Property
Well-ordered	Every subset has unique least element	Stronger than well-founded
Well-founded	Every subset has minimal elements	DCC: no infinite descent
DCC (Artinian)	No infinite descending chains	Same as well-founded
ACC (Noetherian)	No infinite ascending chains	Dual of DCC

Key relationships:

- Well-ordered $\Longrightarrow$ Total order (least elements imply comparability)
- Well-ordered $\Longrightarrow$ Well-founded (every least element is minimal)
- Well-founded $\Leftrightarrow$ Noetherian $\Leftrightarrow$ DCC (equivalent characterizations)
- ACC is independent of DCC (a structure can satisfy one but not the other)
- These concepts are fundamental for proving termination and finiteness properties

Outline

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TODO

- Applications of lattices in:
  - Formal concept analysis
  - Domain theory in computer science
  - Algebraic topology
  - Cryptography (lattice-based cryptography)
- Advanced topics in set theory:
  - Cardinal arithmetic
  - Ordinal numbers
  - Forcing and independence results
  - Large cardinals
- Connections to Boolean algebra (next lecture)
- Applications in formal logic and proof theory

Looking Ahead: Boolean Algebra

The next lecture will explore *Boolean algebra*, which provides the mathematical foundation for:

- Digital circuit design and computer hardware
- Propositional logic and automated reasoning
- Database query optimization
- Formal verification of software and hardware systems

Key topics will include:

- Boolean functions and their representations
- Normal forms (CNF, DNF)
- Minimization techniques (Karnaugh maps, Quine-McCluskey)
- Functional completeness and Post's theorem
- The satisfiability problem (SAT) and its computational complexity

Preview: Formal Logic

Following Boolean algebra, we will study *formal logic*, covering:

- Propositional and predicate logic
- Natural deduction and proof systems
- Completeness and soundness theorems
- Applications to program verification and AI reasoning

This progression from sets $\rightarrow$ relations $\rightarrow$ functions $\rightarrow$ Boolean algebra $\rightarrow$ logic provides a solid foundation for advanced topics in discrete mathematics and computer science.

Binary relations are the bridge between sets and functions — they model how objects *connect*, *organize*, and *interact* in mathematical structures and real-world systems.